جامعة أم القرى
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قسم علوم الحاسب

VYTAL Platform Project

Graduation Project, Part-I I

Computer Science Department

Jamoum University College

Umm Alqura University

Submitted by

Jory Basim Al-Maimani , 444007127

Afnan Saleh Al-Harbi , 444007871

Layan Khalid Al-Harbi , 444006763

Shooq Mohammed Al-Adwani , 444007146

Aryam Fayez Al-Mabadi , 444007571

Project Advisor:

Dr. Ghada AbdulQader Al-Fattni

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Abstract

Athlete health monitoring plays a critical role in optimizing performance, preventing injuries, and supporting timely medical decision-making. However, many existing monitoring approaches primarily focus on presenting physiological measurements and often require manual interpretation of athlete conditions. This project aimed to develop an intelligent athlete health monitoring platform, named VYTAL, that integrates business intelligence, artificial intelligence, and interactive data visualization to support evidence-based athlete health assessment.

The system was developed using Microsoft Power BI, Python, DAX, Power Query, and the Gemini Large Language Model (LLM). Athlete physiological data, including HR, SpO_2, and body temperature, were collected from a publicly available dataset, preprocessed, and organized into a relational analytical model. An evidence-based classification framework was implemented using scientifically validated physiological thresholds to categorize athlete conditions into three risk levels: Green (Normal), Orange (Warning), and Red (Critical). Interactive dashboards were developed to provide real-time analytical insights, athlete comparisons, trend monitoring, and risk visualization. In addition, Python integration and Gemini LLM were utilized to generate automated athlete health interpretations and recommendation-oriented reports.

The developed platform successfully integrated physiological monitoring, automated risk classification, interactive analytics, and AI-assisted interpretation within a unified environment. Testing and evaluation demonstrated stable system performance, accurate classification outputs, responsive dashboard interaction, and reliable AI-generated reports. Usability assessment indicated that the platform was intuitive to use and effectively supported rapid identification of athlete risk conditions, interpretation of physiological data, and analytical decision-making.

The findings demonstrate that the VYTAL platform provides a comprehensive and scalable solution for intelligent athlete health monitoring. By combining evidence-based physiological assessment with AI-powered analytics and visualization, the system enhances situational awareness and supports data-driven decision-making in sports-health environments.

Keywords: Athlete Health Monitoring, Power BI, Artificial Intelligence, Gemini LLM, Sports Analytics, Physiological Monitoring, Risk Classification, Data Visualization.

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Chapter 1

Introduction

1.1 Background and Motivation

The rapid advancement of AI, data analytics, and smart healthcare technologies has significantly transformed the way physiological data can be monitored and interpreted within sports environments. Athletes across different sports disciplines are continuously exposed to intense physical demands that may lead to fatigue, heat stress, oxygen desaturation, and performance deterioration. Consequently, monitoring physiological indicators such as Heart Rate (HR), oxygen saturation (SpO_2), and body temperature has become essential for maintaining athlete safety, optimising performance, and supporting medical decision-making. Despite the increasing availability of athlete physiological data, many existing monitoring approaches remain limited to static measurements or basic visualisations without providing advanced analytical interpretation or dynamic health classification. Coaches and medical staff often require systems capable not only of displaying physiological measurements, but also of analysing athlete conditions, identifying potential risks, and generating understandable insights in real time. This project was motivated by the growing need for an advanced and interactive athlete health monitoring platform that combines AI, data visualization, and evidence-based analytics within a unified intelligent environment. The proposed system integrates Microsoft Power BI, Python, and the Gemini Large Language Model (LLM) to create an AI-powered analytical dashboard capable of transforming raw athlete data into meaningful health insights and automated medical-style interpretations. A major strength of the project lies in the implementation of an AI-driven health classification framework based on scientifically validated physiological thresholds derived from published medical research. Athlete conditions are dynamically classified into green, orange, and red risk levels according to HR, SpO_2, and body temperature measurements. This intelligent classification mechanism enables rapid identification of abnormal physiological patterns and supports proactive athlete monitoring. Furthermore, the project introduces advanced interactive analytical features including player-level analysis, health trend tracking, risk contribution visualisation,multi-player comparison, and AI-generated athlete reports. By integrating advanced analytics with real-time visual interpretation, the system provides a modern decision-support environment that enhances situational awareness for coaches and medical teams. This project demonstrates the potential of integrating artificial intelligence, business intelligence, and evidence-based analytics into a unified sports health monitoring ecosystem capable of supporting proactive athlete assessment, enhancing analytical efficiency, and improving data-driven decision-making within sports healthcare environments.

1.2 Problem Statement

Athletes generate large volumes of physiological data during training sessions and physical activities; however, transforming these measurements into meaningful health insights remains a significant challenge. Traditional monitoring methods often focus primarily on displaying raw numerical values without providing intelligent interpretation, automated classification, or comprehensive analytical support. Additionally, the absence of dynamic risk assessment mechanisms may delay the identification of abnormal physiological conditions such as elevated HR, reduced oxygen saturation, or abnormal body temperature. This limitation reduces the ability of coaches and medical teams to perform efficient athlete monitoring and rapid decision-makin. Therefore, there is a need for an intelligent athlete health monitoring system capable of analyzing physiological data, applying AI-based health classification, visualizing athlete conditions interactively, and generating automated analytical interpretations that support real-time monitoring and informed decision-making.

1.3 Objectives

The primary objectives of this project are:

1. To develop an AI-powered athlete health monitoring and decision-support dashboard using Microsoft Power BI and Python.

2. To implement an evidence-based AI health classification system that categorizes athlete conditions into Green, Orange, and Red risk levels using physiological thresholds derived from scientific literature.

3. To design interactive dashboards and visualizations for athlete monitoring, trend analysis, player comparison, and health risk contribution assessment.

4. To integrate the Gemini LLM for generating automated health interpretations and analytical reports, thereby supporting coaches and medical teams in making informed decisions regarding athlete health and performance.

Chapter 2

Literature Review

2.1 Review of Related Studies

Recent developments in data analytics, AI, and healthcare information systems have significantly enhanced the ability to process and interpret large-scale physiological and clinical data. One of the most prominent developments in this field is the use of interactive dashboards, which enable the visualization of complex datasets in a structured and interpretable manner. Studies have shown that dashboard systems enhance data accessibility, support real-time monitoring, and improve decision-making through visual indicators, performance metrics, and analytical summaries [1], [2]. In healthcare environments, dashboards are widely used to transform raw data into meaningful insights that support operational and clinical decision processes [2], [15].In addition to visualization technologies, AI-based decision support systems have become an essential component of modern analytical environments. These systems utilize machine learning and intelligent algorithms to analyze large datasets, identify patterns, and support decision-making processes. Research indicates that AI-driven systems improve analytical accuracy, reduce manual interpretation effort, and enhance the efficiency of decision-making in complex environments [4], [15]. Furthermore, XAI has emerged as an important advancement, improving transparency and allowing users to understand how AI-generated outputs are derived, which increases trust and usability in decision-support applications [4].Moreover, the integration of LLMs has introduced new capabilities in healthcare analytics and intelligent reporting. LLMs are capable of understanding contextual information, generating human-like explanations, summarizing analytical results, and producing automated reports using NLP techniques. Studies highlight that LLM-based systems improve the interpretability of complex data and enhance user interaction by converting structured datasets into understandable textual insights [5]–[8]. Additionally, LLMs have demonstrated strong potential in supporting clinical decision support systems through automated interpretation and conversational analytical assistance [6], [7].Furthermore, athlete health monitoring has gained significant attention in sports science and performance analysis. Continuous monitoring of physiological indicators is essential for evaluating athlete condition, training response, fatigue levels, and recovery status. Previous research emphasizes the importance of physiological variables such as HR, SpO_2, and body temperature, which serve as key indicators of physiological stress and athlete readiness. These indicators are widely used in sports science to support athlete safety, performance optimization, and health status evaluation.In addition, several studies have explored the relationship between physiological data and athlete performance management. Integrating physiological indicators with analytical models allows for more accurate interpretation of athlete conditions and supports evidence-based decision-making in training and competition environments. Research findings suggest that structured monitoring of physiological responses contributes to early detection of fatigue, injury risk, and performance decline, thereby improving athlete management strategies [10], [11], [14].Overall, the literature indicates significant progress in dashboards, AI-based decision support systems, LLMs, and athlete health monitoring. These advancements collectively contribute to improving data interpretation, analytical efficiency, and decision-making processes in healthcare and sports analytics environments [1]–[15].

2.2 Research Gap and Justification

Although existing studies have demonstrated significant advancements in dashboards, AI, LLMs, and athlete health monitoring, most of these works focus on individual components rather than an integrated analytical environment.Previous research has primarily addressed data visualization, AI-based decision support, and physiological monitoring as separate domains, with limited efforts toward combining these capabilities into a unified system for athlete health assessment.In addition, current athlete monitoring approaches often emphasize the collection and visualization of physiological data without providing sufficient automated interpretation or intelligent classification of athlete conditions. Similarly, while LLMs have been widely explored for generating explanations and supporting healthcare analytics, their integration with real-time physiological monitoring systems and structured health classification frameworks remains limited.Therefore, there is a clear need for an integrated intelligent system that combines real-time physiological monitoring, AI-based health classification, interactive visualization, and LLM-driven analytical interpretation within a single platform. The proposed system addresses this gap by integrating Microsoft Power BI dashboards, Python-based analytical processing, evidence-based physiological thresholds, and LLM capabilities to provide automated insights and support data-driven decision-making in athlete health monitoring.

2.3 Feasibility Study

Technical Feasibility

The proposed system is technically feasible due to the availability and maturity of the required technologies. Power BI provides advanced capabilities for data visualization and interactive dashboard development. Python enables efficient data processing, statistical analysis, and implementation of AI-based classification logic. In addition, cloud-based APIs such as Gemini support LLM integration for generating automated interpretations and analytical reports. The combination of these technologies allows the development of an intelligent and scalable athlete health monitoring system based on existing and well-supported tools.Furthermore, the implementation of AI-based health classification is feasible using rule-based and threshold-driven models derived from established physiological literature. These models can be integrated into the analytical pipeline without requiring high computational complexity.

Financial Feasibility

The project is financially feasible as it primarily relies on software-based tools and platforms that are either free, open-source, or available under academic licensing. Power BI offers educational access, Python libraries are open-source, and LLM APIs such as Gemini can be accessed through scalable and cost-efficient cloud services. Therefore, the system does not require significant financial investment in hardware or infrastructure.

Operational Feasibility

The proposed system is operationally feasible as it is designed to be user-friendly and suitable for non-technical users such as coaches and medical staff. The use of interactive dashboards simplifies the interpretation of complex physiological data, while AI-generated insights enhance understanding and decision-making. Additionally, the system provides automated classification of athlete conditions into risk levels (Green, Orange, Red), which improves usability and supports quick assessment in real-time monitoring environments. The modular design also allows for future expansion and integration of additional analytical features if required.

Chapter 3

Methodology

3.1 Research Methodology

This project adopted a practical applied research methodology combined with a system development approach to design and implement an intelligent athlete health monitoring platform. The methodology focused on integrating data analytics, artificial intelligence, and interactive visualization techniques to analyze athlete physiological data and generate meaningful analytical insights. The project followed a data-driven analytical approach in which athlete physiological measurements were collected, preprocessed, analyzed, classified, and visually interpreted through an AI-powered dashboard environment. The selected methodology was appropriate for the nature of the project because it allowed the integration of multiple technologies, including business intelligence tools, Python-based analytics, and LLMs, within a unified intelligent system. From a development perspective, the project followed an Agile-based iterative development methodology. This approach enabled continuous improvement and gradual enhancement of the dashboard functionalities throughout the development process. Different system modules, including data preprocessing, dashboard visualization, AI-based classification, and Gemini integration, were implemented incrementally and refined through repeated testing and evaluation. The Agile methodology was selected because it provides flexibility, supports iterative development, and allows rapid modification of analytical components and visualization features during implementation. In addition, the methodology facilitated continuous optimization of dashboard design, risk classification logic, and AI-generated interpretation outputs to improve usability and analytical performance.

The overall development process consisted of several major stages:

- Dataset acquisition and preprocessing
- Data cleaning and transformation
- Relational data modeling
- Development of analytical dashboards
- Implementation of AI-based health classification
- Integration of Gemini LLM
- Dashboard testing and validation
- Final visualization optimization and reporting

This methodology enabled the successful development of a scalable and intelligent sports health analytics platform capable of supporting athlete monitoring and data-driven decision-making.

3.2 Tools and Technologies / Data Collection Methods

The project utilized multiple technologies and analytical tools to support data processing, visualization, artificial intelligence integration, and dashboard development.

Figure 1VYTAL System Development and Data Processing Workflow

Programming Languages and Technologies

The following technologies and tools were used during project development:

Figure 2Technology Stack Used for System Development

Technology / Tool	Purpose
Python	Data analysis, AI integration, and report generation
Microsoft Power BI	Interactive dashboard development and visualization
Gemini LLM	AI-generated athlete health interpretation
DAX (Data Analysis Expressions)	Calculated measures and dynamic analytics
Kaggle Dataset	Source of athlete physiological data
Power Query	Data cleaning and transformation

Table 1Technologies and Tools Used in System Development

AI-Based Health Classification

One of the core features of the project was the implementation of an AI-driven health classification framework designed to evaluate athlete physiological conditions dynamically. The system classified athlete health status into three primary risk categories:

- Green (Safe)
- Orange (Warning)
- Red (Critical)

The classification framework was developed using scientifically validated physiological thresholds obtained from published medical literature [17]–[19]. These thresholds were implemented as rule-based classification logic within Microsoft Power BI using DAX calculations. For each athlete, the system automatically evaluates physiological measurements, including HR [17], SpO_2 [18], and body temperature [19], against predefined threshold ranges. Based on the evaluation results, the athlete is dynamically assigned to one of three health-risk categories: Green (Normal), Orange (Warning), or Red (Critical). The generated classification results are then used to update dashboard indicators, risk alerts, KPI cards, analytical visualizations, and AI-generated interpretation outputs, enabling automated identification of athlete risk conditions and supporting proactive monitoring.

The implemented physiological classification thresholds were as follows:

Heart Rate(HR as Percentage of HR Peak)

Risk Level	Threshold	Interpretation
Green (Normal)	Less than 85% of HR Peak	Normal physiological performance with no significant fatigue
Orange (Warning)	85% to less than 93% of HR Peak	Early signs of physical stress requiring monitoring
Red (Critical)	93% of HR Peak or higher	High physiological stress and elevated risk condition

Core Body Temperature

Risk Level	Threshold	Interpretation
Green (Normal)	36.0°C – 38.0°C	Normal thermal regulation
Orange (Warning)	38.1°C – 39.5°C	Heat stress and early risk indication
Red (Critical)	39.6°C or higher	High-risk heat stress condition

Oxygen Saturation (SpO_2)

Risk Level	Threshold	Interpretation
Green (Normal)	95% – 100%	Normal oxygen saturation
Orange (Warning)	90% – 94%	Mild oxygen reduction requiring monitoring
Red (Critical)	Less than 90%	Hypoxia and critical oxygen deficiency

Table 2Physiological Thresholds Used for Athlete Health Risk Classification

The implemented AI-based classification framework enhanced the analytical capabilities of the system by enabling automated physiological risk assessment, dynamic health status visualization, and intelligent athlete condition monitoring within the dashboard environment.

Gemini LLM Integration

The Gemini LLM was integrated into the system using Python to generate automated analytical interpretations based on athlete physiological readings. The AI model analyzed HR, SpO_2, body temperature, session duration, and alert indicators to generate concise structured reports containing:

- Athlete status
- Key findings
- Recommendations

The generated reports enhanced interpretability by transforming numerical physiological measurements into understandable analytical insights for coaches and medical teams.

3.3 Risk Management

Several potential risks were considered during project development to ensure system stability, analytical consistency, and implementation efficiency.

Risk	Likelihood	Impact	Mitigation Strategy
Data inconsistency or duplicate records	Medium	High	Applying comprehensive data cleaning, preprocessing, and duplicate removal techniques before analysis
Incorrect analytical calculations or dashboard measures	Medium	High	Validating DAX measures, reviewing analytical outputs, and performing repeated testing of calculations
AI-generated interpretation inaccuracies	Medium	Medium	Using controlled prompting techniques and structured AI response generation to improve consistency
Dashboard performance or visualization delays	Low	Medium	Optimizing data modeling, reducing unnecessary calculations, and improving visualization efficiency
Integration challenges between Python, Power BI, and Gemini LLM	Medium	Medium	Conducting incremental integration testing throughout the development process
Misclassification of athlete physiological conditions	Low	High	Implementing evidence-based physiological thresholds derived from published scientific literature
Inconsistent visualization or user interaction behavior	Low	Medium	Reviewing dashboard usability, interactivity, and visualization consistency during testing
Data transformation or preprocessing errors	Medium	Medium	Verifying transformed datasets and validating preprocessing procedures before implementation

Table 3Project Risk Management

3.4 Quality Assurance Plan / Validation and Reliability

Several validation and quality assurance procedures were implemented to ensure the accuracy, reliability, and consistency of the developed system. The dataset underwent preprocessing and validation procedures to reduce inconsistencies and improve data quality before analysis. In addition, analytical measures and calculated fields within Power BI were repeatedly tested to verify the correctness of dashboard outputs and risk calculations.

The AI-based health classification framework was validated using scientifically supported physiological threshold values derived from published medical studies. This evidence-based approach enhanced the reliability of athlete risk classification and ensured consistency between physiological measurements and generated health status categories.

Furthermore, the Gemini LLM integration was evaluated through repeated testing of different athlete scenarios to ensure that the generated reports remained concise, structured, and aligned with the provided physiological data. Controlled prompting techniques were implemented to maintain consistency in AI-generated outputs.

The dashboard interface and visualization components were also validated through systematic functional and visual testing procedures. Multiple athlete scenarios representing Green, Orange, and Red risk conditions were applied to verify that dashboard components responded correctly to changing physiological measurements. Interactive features such as filters, slicers, athlete selection controls, and cross-page navigation were tested to ensure synchronized behavior across all dashboard pages. KPI cards, trend charts, risk indicators, comparison visuals, and diagnostic panels were validated against the underlying dataset and DAX calculations to confirm accurate data representation. In addition, visualization layouts, color-coded status indicators, and alert banners were evaluated to ensure consistency, readability, and clear interpretation of athlete health conditions.

These validation procedures contributed to the development of a reliable and effective intelligent athlete health monitoring platform.

Chapter 4

Project Requirements & Analysis

4.1 Stakeholder Identification

The developed athlete health monitoring platform involves several stakeholders who interact with the system in different ways depending on their roles and objectives. Identifying stakeholders was essential for understanding system requirements, defining dashboard functionalities, and ensuring that the platform supports effective athlete monitoring and analytical decision-making.

Coaches

Coaches represent one of the primary stakeholders of the system. They utilize the dashboard to monitor athlete physiological conditions, evaluate performance-related indicators, identify potential fatigue or stress conditions, and support training-related decisions. The interactive visualizations and AI-generated insights assist coaches in tracking athlete status efficiently.

Medical Teams and Sports Health Specialists

Medical teams and sports health professionals use the system to observe athlete physiological measurements and identify abnormal health patterns that may require attention. The AI-based health classification framework and automated analytical interpretations support rapid understanding of athlete conditions and improve monitoring efficiency.

Athletes

Athletes represent another important stakeholder group. The platform provides clear visual representations of physiological conditions, enabling athletes to better understand their health status, performance trends, and potential risk levels during training activities.

Data Analysts and Researchers

Data analysts and researchers can utilize the platform to analyze physiological trends, evaluate athlete performance data, and study relationships between physiological indicators and athlete conditions. The dashboard provides analytical capabilities that support data-driven sports analysis.

Project Developers and System Administrators

Developers and system administrators are responsible for system implementation, dashboard maintenance, AI integration, and analytical optimization. They ensure system functionality, data consistency, and visualization accuracy throughout the development lifecycle.

4.2 Methods of Requirements Gathering / Data Needs and Sources

The project requirements were identified through analytical evaluation of athlete monitoring needs, review of existing sports health analytics approaches, and examination of physiological monitoring practices used in sports environments. The system was designed to address the need for intelligent athlete monitoring, dynamic health classification, and AI-powered analytical interpretation.

Data Needs

The project required physiological athlete data capable of supporting analytical monitoring and health classification processes. The primary data requirements included:

- Heart Rate
- Oxygen Saturation
- Body Temperature
- Session Duration
- Athlete Alert Indicators

These physiological indicators were selected because of their importance in evaluating athlete health conditions, fatigue levels, and physiological stress during physical activity.

Data Source

The dataset used in the project was obtained from Kaggle. The dataset contained structured physiological information related to athlete monitoring and sports performance analysis.

Before implementation, the dataset underwent multiple preprocessing stages including:

- Data cleaning
- Duplicate removal
- Data transformation
- Data restructuring
- Relational organization for analytical modeling

These preprocessing procedures improved data quality and ensured compatibility with Power BI analytical workflows.

Scientific Literature and Threshold Sources

To support the AI-based health classification framework, scientifically validated physiological thresholds were collected from published medical literature related to:

- HR peak percentages
- Oxygen saturation ranges
- Core body temperature levels

These evidence-based thresholds formed the foundation of the athlete risk classification system implemented within the dashboard.

4.3 Functional Requirements

The developed system provides several functional capabilities designed to support intelligent athlete health monitoring and analytical decision-making.

Athlete Monitoring Functions

- Display athlete physiological measurements including HR, SpO_2, and body temperature
- Monitor athlete health status dynamically
- Track athlete physiological trends over time

AI-Based Health Classification

- Classify athlete conditions into Green, Orange, and Red risk levels
- Detect abnormal physiological conditions automatically
- Visualize athlete risk status dynamically

Dashboard Analytics and Visualization

- Provide interactive dashboard visualizations
- Support player-level analysis
- Enable multi-player comparison
- Display health contribution and risk metrics
- Present analytical charts and performance indicators

AI-Generated Interpretation

- Generate automated athlete health interpretations using Gemini LLM
- Produce structured analytical reports
- Generate recommendations based on physiological readings

Data Management and Processing

- Process and transform athlete physiological data
- Support relational data modeling
- Calculate dynamic analytical measures using DAX

User Interaction Features

- Filter dashboard results dynamically
- Navigate between analytical dashboard pages
- Interact with visual analytical components

4.4 Non-Functional Requirements

The project includes several non-functional requirements to ensure system quality, usability, reliability, and analytical efficiency.

Performance Requirements

- The dashboard should process and display athlete data efficiently.
- Analytical visualizations should update dynamically with minimal delay.
- AI-generated interpretations should be generated within an acceptable response time.

Usability Requirements

- The dashboard interface should be interactive and user-friendly.
- Visualizations should present physiological information clearly and consistently.
- Users should be able to navigate between dashboard sections easily.

Reliability Requirements

- The analytical calculations and risk classifications should remain consistent and accurate.
- AI-generated outputs should align with the provided physiological measurements.
- Data preprocessing procedures should improve data integrity and consistency.

Scalability Requirements

- The system architecture should support future integration of additional physiological indicators.
- The platform should support expansion toward real-time athlete monitoring environments.

Security and Data Handling Requirements

- Athlete data should be handled within a controlled analytical environment.

Access to dashboard components and analytical outputs should remain organized and structured.

4.5 Constraints

Several constraints were considered during the development of the project. The implemented system focused primarily on selected physiological indicators including HR, SpO_2, and body temperature in order to maintain analytical clarity and dashboard efficiency. The project was developed using a structured dataset obtained from Kaggle to simulate athlete monitoring and analytical workflows within the dashboard environment. In addition, the AI-generated interpretations were designed to support analytical understanding and monitoring assistance rather than clinical diagnosis. The project scope focused specifically on intelligent analytics, interactive visualization, and AI-driven classification functionalities within a business intelligence environment. Future extensions may include additional physiological parameters, wearable device integration, and advanced predictive modeling capabilities.

Chapter 5

System Design

5.1 Overview of Design Approach

The design approach of this project was centered on developing an intelligent, interactive, and data-driven athlete health monitoring platform capable of transforming physiological measurements into meaningful analytical insights. The overall design philosophy focused on combining artificial intelligence, business intelligence, and sports health analytics within a unified environment that supports efficient athlete monitoring and data-driven decision-making.

The system was designed with an emphasis on analytical clarity, real-time visualization, intelligent risk assessment, and user interaction. Rather than presenting physiological measurements as isolated numerical values, the platform was developed to provide dynamic interpretation, AI-based classification, and visually intuitive representations of athlete health conditions.

A modular and scalable design strategy was adopted during system development. The architecture separated the project into interconnected analytical components, including:

- Data preprocessing and transformation
- Relational data modeling
- Interactive dashboard visualization
- AI-based health classification
- Gemini LLM integration
- Automated report generation

This modular approach improved maintainability, scalability, and flexibility while allowing different system components to operate cohesively within the dashboard environment.

The dashboard interface was designed using a user-centered analytical approach to ensure that physiological information could be interpreted quickly and efficiently by coaches, medical teams, and analysts. Visual components such as dynamic charts, risk indicators, status cards, trend analysis visuals, and comparative analytics were implemented to improve situational awareness and support rapid understanding of athlete conditions.

A key aspect of the design philosophy was the implementation of an AI-driven health classification framework based on evidence-based physiological thresholds derived from scientific literature. Athlete conditions were dynamically categorized into Green, Orange, and Red risk levels according to HR, SpO_2, and body temperature measurements. This classification mechanism enabled automated risk identification and enhanced the intelligence of the analytical system.

In addition, the integration of the Gemini LLM introduced an advanced analytical layer capable of generating structured AI-powered athlete health interpretations and recommendations. This design decision transformed the platform from a traditional visualization dashboard into an intelligent analytical decision-support system capable of producing understandable medical-style insights from complex physiological data.

The overall system design also prioritized consistency, responsiveness, and visual simplicity. Clean dashboard layouts, organized analytical sections, interactive filtering capabilities, and dynamic visual feedback were implemented to enhance usability and analytical efficiency.

Through the integration of artificial intelligence, interactive analytics, and evidence-based classification methods, the final design achieved a scalable and intelligent sports health monitoring ecosystem capable of supporting proactive athlete assessment and advanced analytical monitoring.

5.2 Data Flow Analysis

Figure 3 presents the Context-Level DFD, illustrating the interaction between the VYTAL Athlete Health Monitoring System and its primary external entities, including the Coach, Medical Team, Data Analyst, Athlete, and System Administrator.

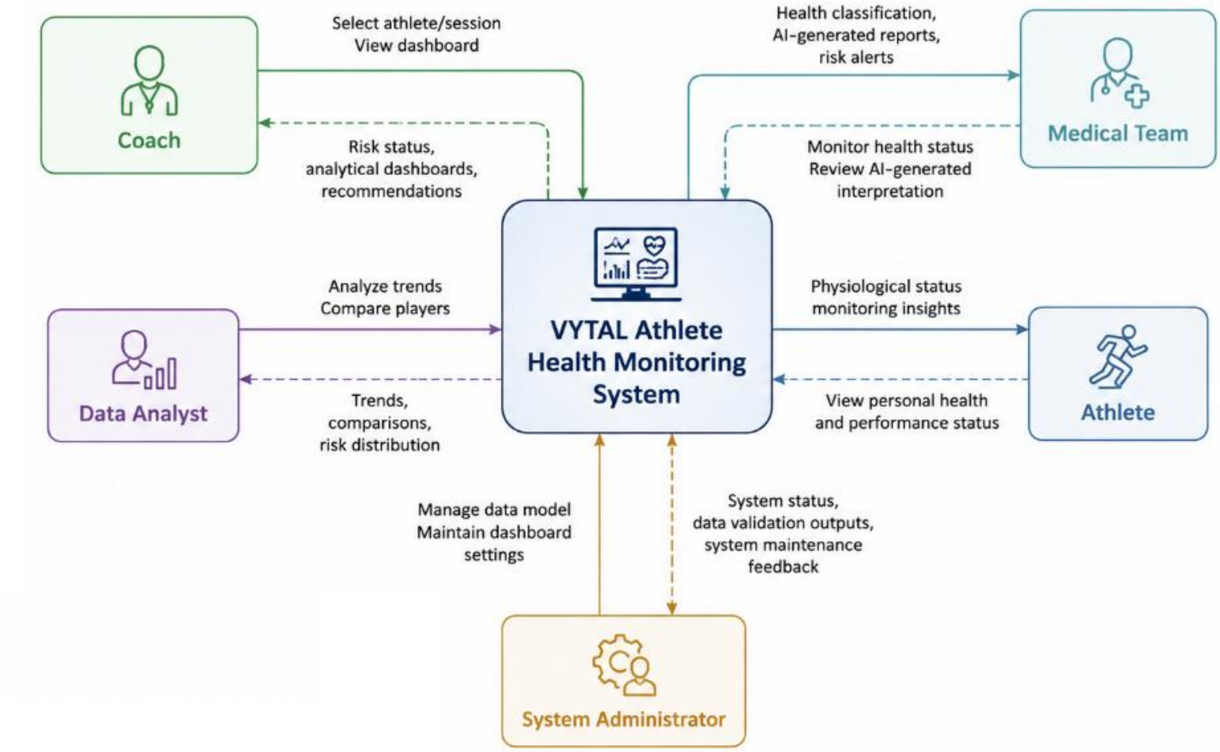

Figure 3Context-Level DFD of the VYTAL Athlete Health Monitoring System

Figure 4 presents the Level 1 DFD, providing a detailed representation of the internal system workflow, including data collection, preprocessing, analytical modeling, physiological risk classification, dashboard visualization, and AI-powered interpretation generation through Python and Gemini integration.

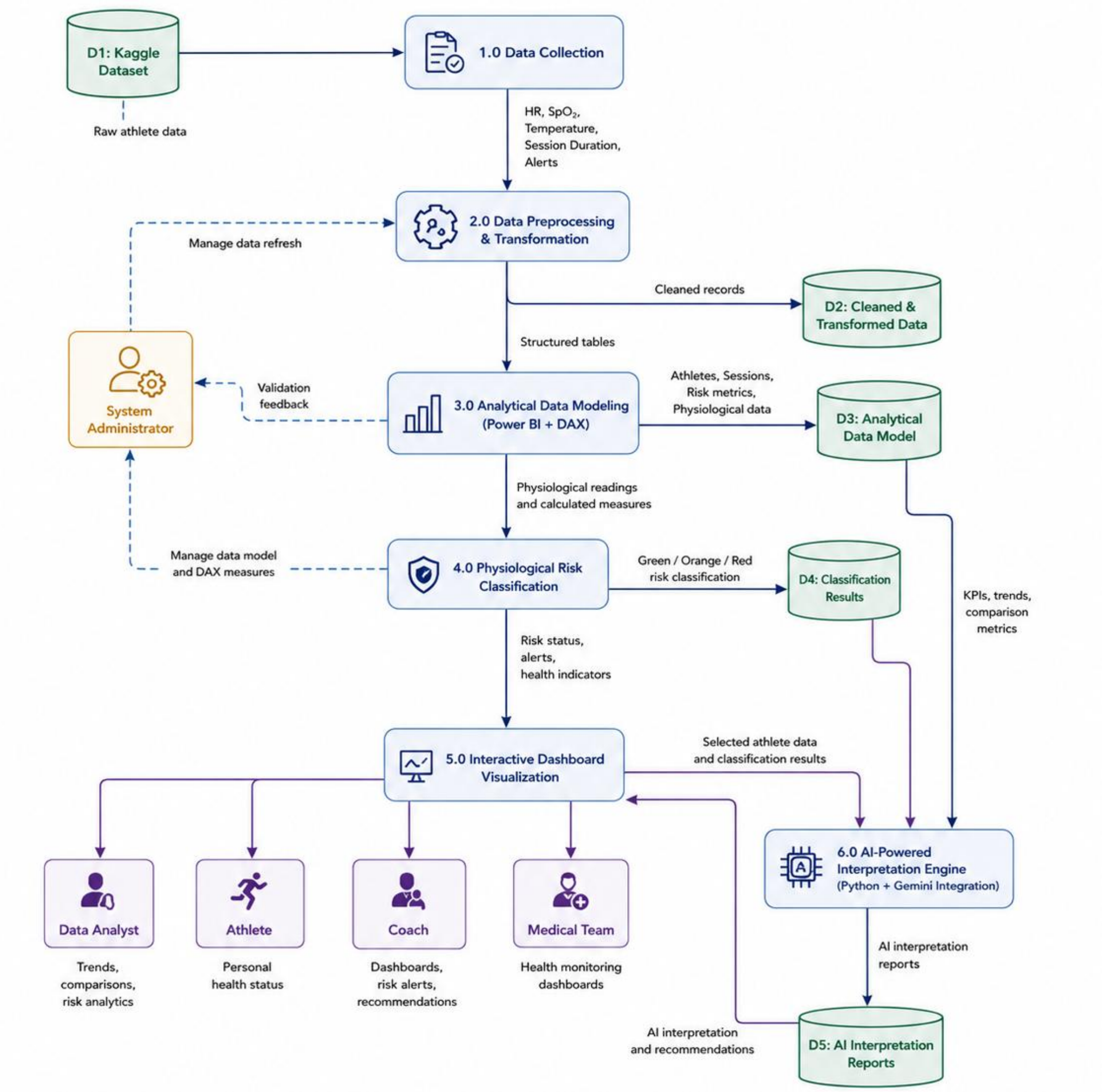

Figure 4Level 1 DFD of the VYTAL Athlete Health Monitoring System

5.3 System Architecture

Figure 5 illustrates the overall architecture of the proposed VYTAL Athlete Health Monitoring System. The architecture demonstrates the flow of athlete physiological data from data collection and preprocessing through analytical modeling, risk classification, dashboard visualization, Python integration, Gemini-based interpretation, and AI-powered report generation. The figure also highlights the interaction between system users and the analytical components within the Power BI environment.

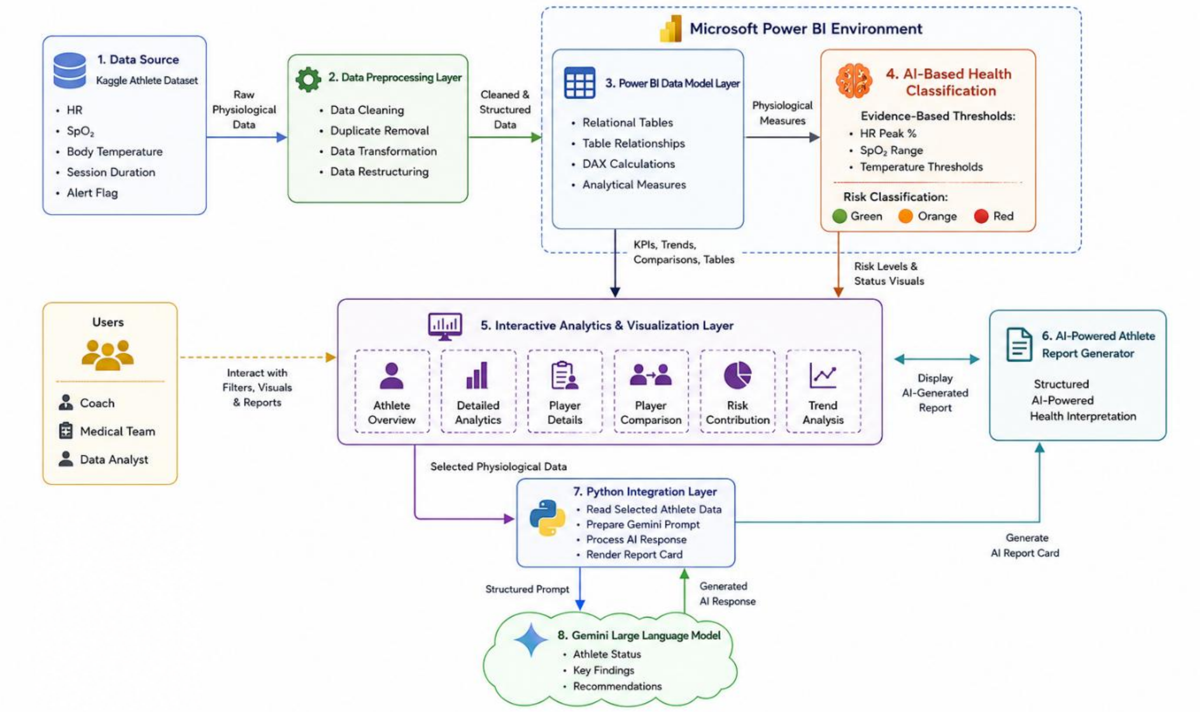

Figure 5System Architecture of the VYTAL Athlete Health Monitoring Platform

5.4 Use Case Analysis

The VYTAL platform was designed to support multiple user interactions within an intelligent analytical environment. The system enables users to monitor athlete physiological conditions, analyze health-related indicators, generate AI-powered interpretations, and visualize analytical insights through interactive dashboard components.

The primary actors interacting with the system include coaches, medical teams, athletes, analysts, and system administrators. Each actor interacts with the platform according to specific operational and analytical requirements.

5.4.1 Main Actors

Actor	Role
Coach	Monitors athlete conditions, evaluates performance indicators, and reviews health risks
Medical Team	Reviews physiological measurements and AI-generated health interpretations
Athlete	Views personal physiological status and performance-related insights
Data Analyst	Analyzes athlete trends, comparisons, and dashboard analytics
System Administrator	Maintains dashboard functionality, manages data integration, and monitors system performance

Table 4Main Actors

5.4.2 Main Use Cases

Athlete Health Monitoring

The system allows coaches and medical teams to monitor athlete physiological indicators including HR, SpO_2, and body temperature through interactive dashboards and visual analytics.

AI-Based Health Classification

The platform automatically classifies athlete conditions into Green, Orange, and Red risk levels using evidence-based physiological thresholds and analytical calculations.

Player-Level Analysis

Users can select individual athletes to review detailed physiological measurements, health trends, and risk indicators.

Multi-Player Comparison

The system enables comparison between multiple athletes to evaluate physiological conditions, performance metrics, and overall risk distribution.

AI-Generated Health Interpretation

The Gemini LLM generates structured analytical reports based on athlete physiological data, including:

- Athlete status
- Key findings
- Recommendations

Trend and Risk Visualization

The platform visualizes athlete physiological trends, health status distribution, and risk contribution metrics using interactive charts and analytical visuals.

Dashboard Interaction and Filtering

Users can interact dynamically with the dashboard through filters, slicers, and navigation components to customize analytical views and explore specific athlete data.

Data Processing and Analytics

The system processes physiological datasets, calculates analytical measures, and updates dashboard visualizations dynamically using Power BI and Python integration.

Use Case Relationships

The implemented use cases operate together within a unified analytical workflow. Athlete physiological data is processed and analyzed by the system, classified using AI-based risk assessment, visualized interactively through dashboard components, and interpreted using Gemini-generated analytical reports. This integrated workflow supports efficient athlete monitoring and intelligent decision-making within sports environments.

5.5 Entity-Relationship (ER) Analysis

Figure 6 ERD of the VYTAL Athlete Health Monitoring System. The diagram illustrates the relationships between athlete records, monitoring sessions, physiological readings, risk classification outputs, diagnostic metrics, risk contribution analysis, latest player status summaries, and AI-generated reports. It also shows the primary and foreign key relationships used to support analytical processing, physiological risk classification, dashboard monitoring, and AI-powered interpretation generation.

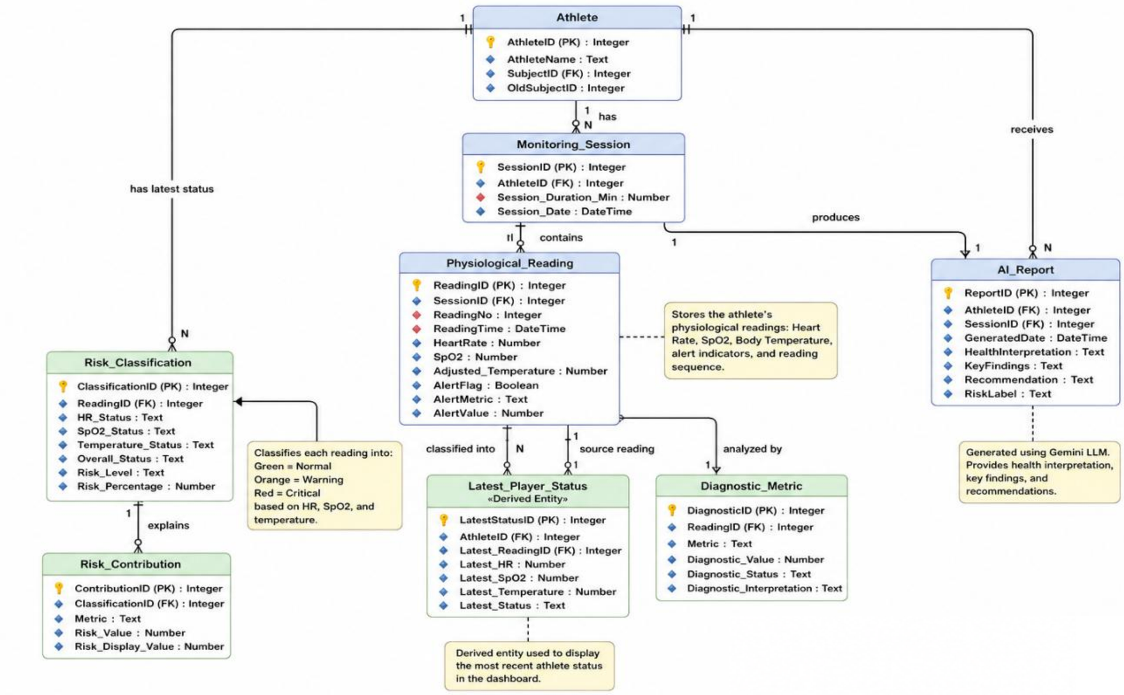

Figure 6ERD of the VYTAL Athlete Health Monitoring System

5.6 System Use Case Diagram

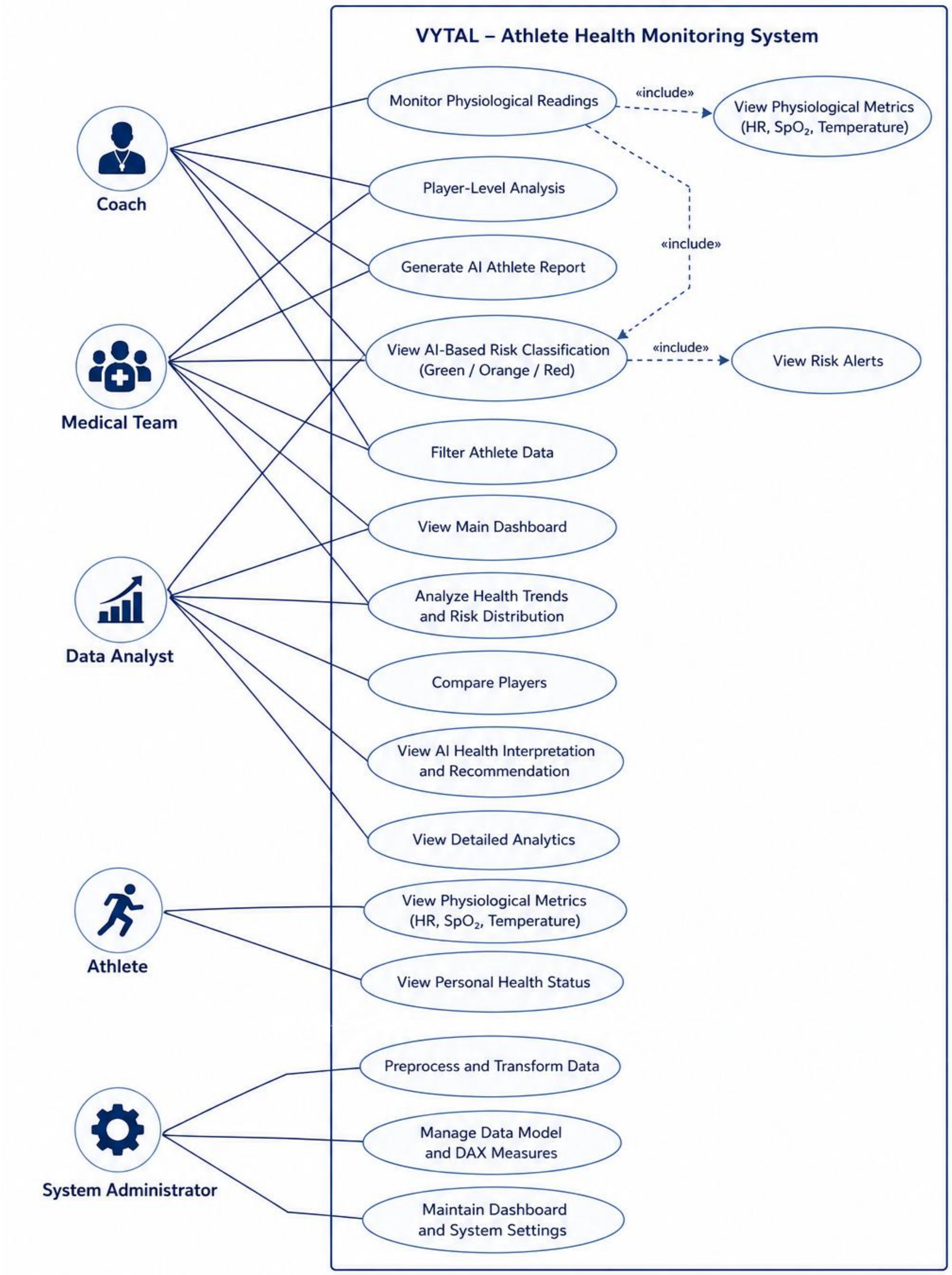

Figure 7Use Case Diagram of the VYTAL Athlete Health Monitoring System

5.7 System Use Case Description

This section provides detailed descriptions of the major system use cases implemented in the VYTAL Athlete Health Monitoring Platform. Each use case describes the interaction flow between system actors and the analytical dashboard environment.

Table 5Detailed Use Case Descriptions of the VYTAL Athlete Health Monitoring System

Use Case 1: Monitor Physiological Readings

Item	Description
Use Case Name	Monitor Physiological Readings
Primary Actors	Coach, Medical Team
Description	Allows users to monitor athlete physiological indicators through interactive dashboard visuals and analytical metrics.
Preconditions	Physiological dataset is loaded and dashboard is active.
Main Flow	1. User opens dashboard. 2. System loads athlete data. 3. User selects athlete or session. 4. Dashboard displays HR, SpO_2, temperature, and session indicators. 5. System updates charts and KPI visuals dynamically.
Alternative Flow	User applies filters to display specific athlete groups or sessions.
Exception Flow	If data is unavailable, the system displays incomplete data warning messages.
Postconditions	Athlete physiological measurements are visualized successfully.

Use Case 2: AI-Based Risk Classification

Item	Description
Use Case Name	AI-Based Risk Classification
Primary Actors	Coach, Medical Team
Description	Automatically classifies athlete health status into Green, Orange, or Red categories using evidence-based thresholds.
Preconditions	Physiological readings are available.
Main Flow	1. System receives HR, SpO_2, and temperature values. 2. Threshold rules are evaluated. 3. Risk percentage is calculated. 4. Athlete status is classified. 5. Dashboard risk visuals are updated.
Alternative Flow	System recalculates classification when filters or new readings are selected.
Exception Flow	Invalid physiological values trigger a validation warning.
Postconditions	Athlete risk status is displayed on the dashboard.

Use Case 3: Player-Level Analysis

Item	Description
Use Case Name	Player-Level Analysis
Primary Actors	Coach, Medical Team, Data Analyst
Description	Allows users to inspect detailed physiological measurements and trends for a selected athlete.
Preconditions	Athlete data exists in the dashboard dataset.
Main Flow	1. User selects an athlete. 2. System retrieves athlete readings. 3. Dashboard displays detailed metrics and trends. 4. User reviews analytical indicators and risk distribution.
Alternative Flow	User changes session filters to display different athlete conditions.
Exception Flow	If selected athlete data is missing, the dashboard displays unavailable data notifications.
Postconditions	Detailed athlete analysis is visualized successfully.

Use Case 4: Compare Players

Item	Description
Use Case Name	Compare Players
Primary Actors	Coach, Data Analyst
Description	Enables comparison between multiple athletes using physiological indicators and analytical metrics.
Preconditions	Multiple athlete records are available.
Main Flow	1. User selects Player A and Player B. 2. System retrieves both athlete datasets. 3. Dashboard generates comparison visuals. 4. User reviews risk differences and physiological trends.
Alternative Flow	User compares athletes from different sessions or teams.
Exception Flow	If comparison data is incomplete, the system displays partial comparison results.
Postconditions	Comparative analytics are displayed successfully.

Use Case 5: Generate AI Athlete Report

Item	Description
Use Case Name	Generate AI Athlete Report
Primary Actors	Coach, Medical Team
Description	Generates AI-powered athlete health interpretations using Gemini LLM integration.
Preconditions	Selected athlete readings are available and Gemini integration is active.
Main Flow	1. User requests AI report. 2. Python layer prepares structured prompt. 3. Prompt is sent to Gemini LLM. 4. AI response is generated. 5. System formats AI interpretation report. 6. Dashboard displays AI-generated report.
Alternative Flow	User regenerates report after changing athlete filters.
Exception Flow	If Gemini API is unavailable, the system displays report generation failure message.
Postconditions	AI-generated interpretation and recommendations are displayed successfully.

Use Case 6: Analyze Health Trends and Risk Distribution

Item	Description
Use Case Name	Analyze Health Trends and Risk Distribution
Primary Actors	Coach, Medical Team, Data Analyst
Description	Allows users to visualize health trends, risk percentages, and athlete distribution analytics.
Preconditions	Dashboard analytical measures are calculated successfully.
Main Flow	1. System calculates trend measures. 2. Dashboard updates trend visuals and risk charts. 3. User reviews analytical insights and performance patterns.
Alternative Flow	User filters trends by athlete, session, or team.
Exception Flow	Missing analytical measures result in incomplete visualizations.
Postconditions	Trend analysis visuals are displayed successfully.

Use Case 7: Manage Data Model and Dashboard Settings

Item	Description
Use Case Name	Manage Data Model and Dashboard Settings
Primary Actors	System Administrator
Description	Allows administrators to maintain dashboard functionality, manage datasets, and update analytical measures.
Preconditions	Administrator access privileges are granted.
Main Flow	1. Administrator updates dataset connections.2. System refreshes relational tables and DAX measures.3. Dashboard configuration is updated.4. System validates analytical calculations.
Alternative Flow	Administrator modifies dashboard settings or data relationships.
Exception Flow	Invalid data model changes trigger configuration warnings.
Postconditions	Dashboard environment is updated successfully.

5.8 System Class Diagram

Figure 8 presents the UML Class Diagram of the VYTAL Athlete Health Monitoring System. The diagram illustrates the main system classes, their attributes, methods, and relationships involved in athlete monitoring, physiological data processing, health classification, dashboard visualization, diagnostic analysis, risk contribution assessment, and AI-powered report generation. It demonstrates how analytical, visualization, and AI components interact within the overall system architecture.

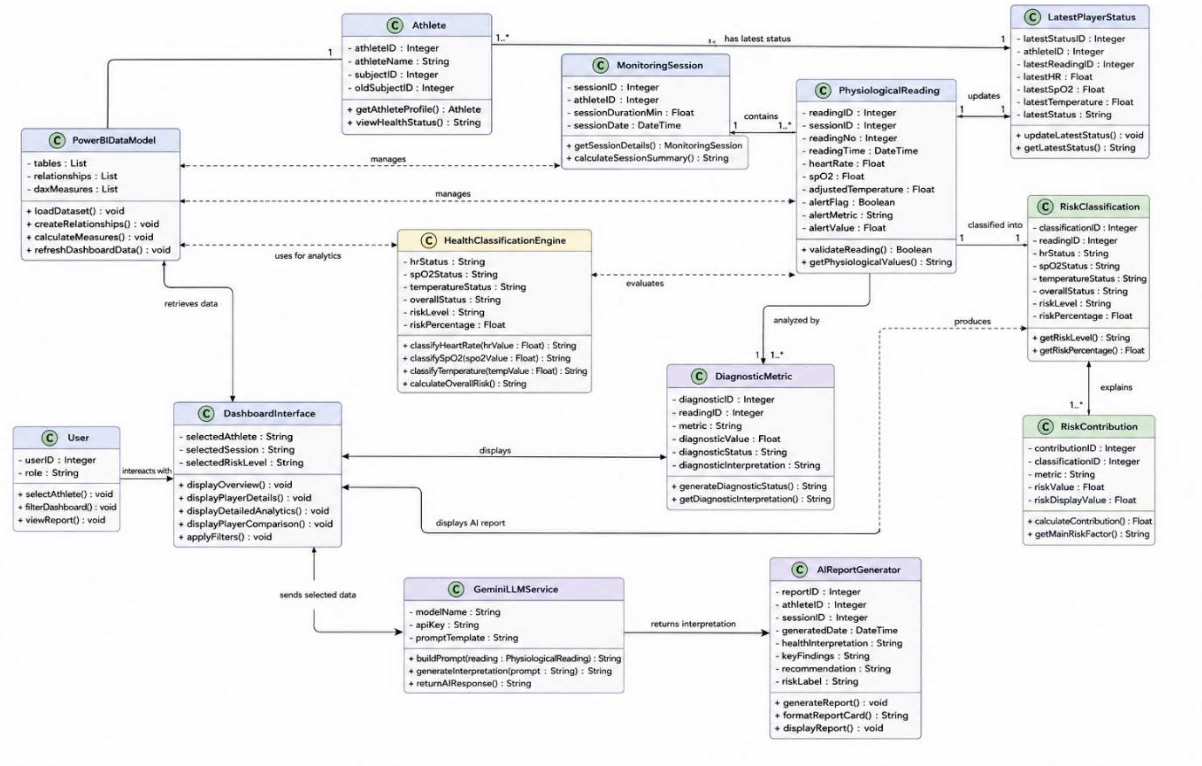

Figure 8UML Class Diagram of the VYTAL Athlete Health Monitoring System

5.9 System Interaction Diagram

Figure 9 presents the system interaction sequence diagram of the VYTAL Athlete Health Monitoring System. The diagram illustrates the sequence of interactions between users, Power BI analytical components, the health classification engine, Python integration layer, Gemini LLM service, and the AI report generation module. It demonstrates how physiological data is processed, classified, visualized, and transformed into AI-generated interpretations and reports through the integrated monitoring workflow.

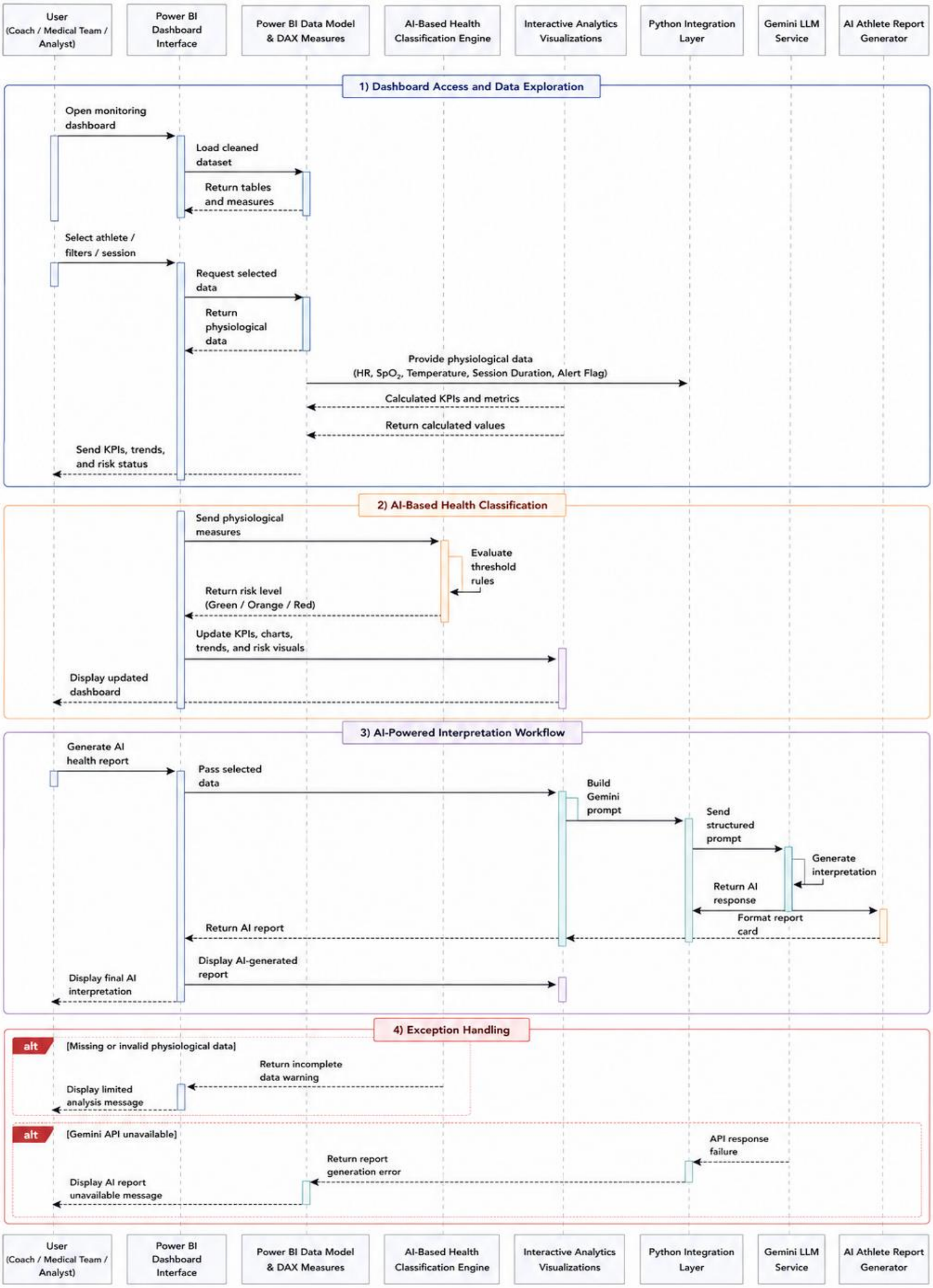

Figure 9System Interaction Sequence Diagram of the VYTAL Athlete Health Monitoring System

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5.10 System Activity Diagram

Figure 10 presents the activity diagram of the VYTAL Athlete Health Monitoring System. The diagram illustrates the operational workflow of the platform, beginning with data loading, preprocessing, analytical modeling, and physiological risk classification, followed by dashboard visualization, AI-powered interpretation generation, athlete comparison, and decision-support activities. It also demonstrates how different risk conditions trigger corresponding monitoring actions and recommendations.

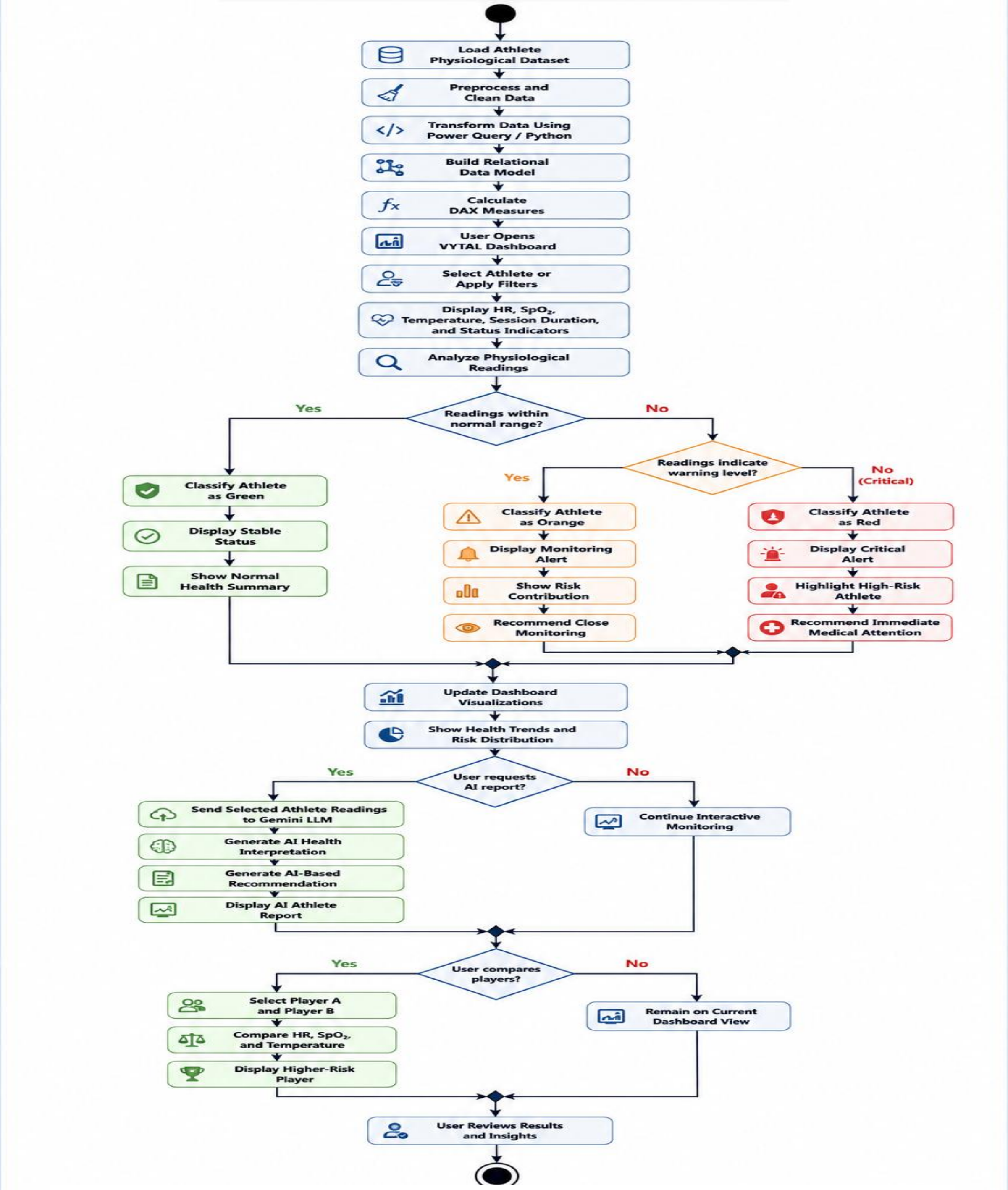

Figure 10System Activity Diagram of the VYTAL Athlete Health Monitoring System

5.11 Front-End / User Interface Design

The front-end design of the VYTAL Athlete Health Monitoring Platform was developed to provide a clear, interactive, and clinically oriented analytical environment for monitoring athlete physiological conditions. The dashboard interface was designed with emphasis on usability, rapid information interpretation, visual consistency, and efficient analytical navigation. The overall interface combines real-time physiological monitoring with AI-powered health interpretation to support decision-making for coaches, medical teams, and analysts.

The user interface was implemented using Microsoft Power BI interactive dashboard components, supported by custom visual formatting, analytical layouts, KPI indicators, slicers, and AI-generated reporting integration. The dashboard structure was divided into multiple analytical pages, where each page focuses on a specific operational or analytical objective within the monitoring workflow.

Dashboard Layout Design

The interface layout follows a modular dashboard architecture that organizes analytical content into visually separated and function-oriented sections. The dashboard was designed to minimize cognitive overload while maximizing the visibility of critical physiological indicators, analytical insights, and athlete risk conditions.

The main dashboard structure includes:

- Top navigation and monitoring header
- Dynamic athlete status indicators
- KPI summary cards
- Trend analysis visualizations
- Risk classification visuals
- Interactive analytical filters and slicers
- AI-powered medical interpretation sections
- Comparative analytics views

The placement of interface components follows a hierarchical top-to-bottom analytical workflow, allowing users to transition naturally from high-level monitoring indicators toward detailed athlete-level analysis and AI-generated health insights.

The platform was divided into multiple analytical pages, where each page was designed to address a specific operational or analytical objective within the athlete monitoring workflow. The implemented dashboard modules include:

- Athlete Health Overview
- Detailed Analytics
- Player Details
- Comparison Players
- Smart Medical Interpretation
- AI Athlete Report Generator

Each dashboard page was structured using organized visual sections to improve navigation efficiency, analytical readability, and information accessibility. Critical physiological metrics and risk indicators were positioned prominently to enhance situational awareness and support rapid interpretation of athlete conditions.

The interface layout incorporates multiple visualization and interaction components, including:

- Summary metric cards
- Interactive charts and graphs
- Risk classification indicators
- Comparative analytical visuals
- Dynamic data tables
- Trend analysis components
- AI-generated interpretation panels

This structured design approach improved dashboard usability, reduced unnecessary visual complexity, and enabled users to interact efficiently with physiological data, analytical insights, and AI-generated health interpretations within a unified monitoring environment.

Athlete Health Overview Interface

The Athlete Health Overview page was designed as the primary monitoring dashboard for coaches and medical teams. This page provides a summarized view of the overall physiological condition of monitored athletes.

Figure 11Athlete Health Overview Dashboard Interface

The interface includes:

- Total active athlete count
- Team risk percentage
- Highest-risk athlete identification
- Critical athlete indicators
- Physiological stress trend visualization
- Risk distribution donut chart
- Navigation tabs for dashboard modules

Large KPI cards and alert banners were intentionally positioned at the top of the interface to prioritize critical health information visibility. Risk colors were implemented consistently throughout the dashboard using:

- Green → Stable condition
- Orange → Monitoring required
- Red → Critical condition

This color mapping improves rapid visual interpretation and supports faster decision-making during athlete monitoring.

Detailed Analytics Interface

The Detailed Analytics page focuses on analytical exploration and population-level physiological evaluation. The layout combines charts, tables, gauges, and categorical indicators to provide a deeper understanding of athlete conditions.

Figure 12Detailed Analytics Dashboard Interface

Implemented interface components include:

- Athlete risk distribution charts
- Health indicator scatter visualizations
- Monitoring coverage gauges
- Athlete data tables
- Risk category selection panels
- Team physiological risk gauges

Interactive slicers and dropdown components were added to allow users to dynamically filter athlete groups and analytical categories without reloading the dashboard environment.

The interface design emphasizes analytical readability through:

- Balanced spacing between visual components
- Consistent typography hierarchy
- Minimal visual clutter
- High-contrast KPI displays
- Structured alignment of analytical panels

Player Details Interface

The Player Details page was designed to support athlete-specific physiological monitoring and individual health assessment. The interface provides detailed insights into the selected athlete's current physiological condition through dynamic indicators, monitoring visuals, and trend-based analysis.

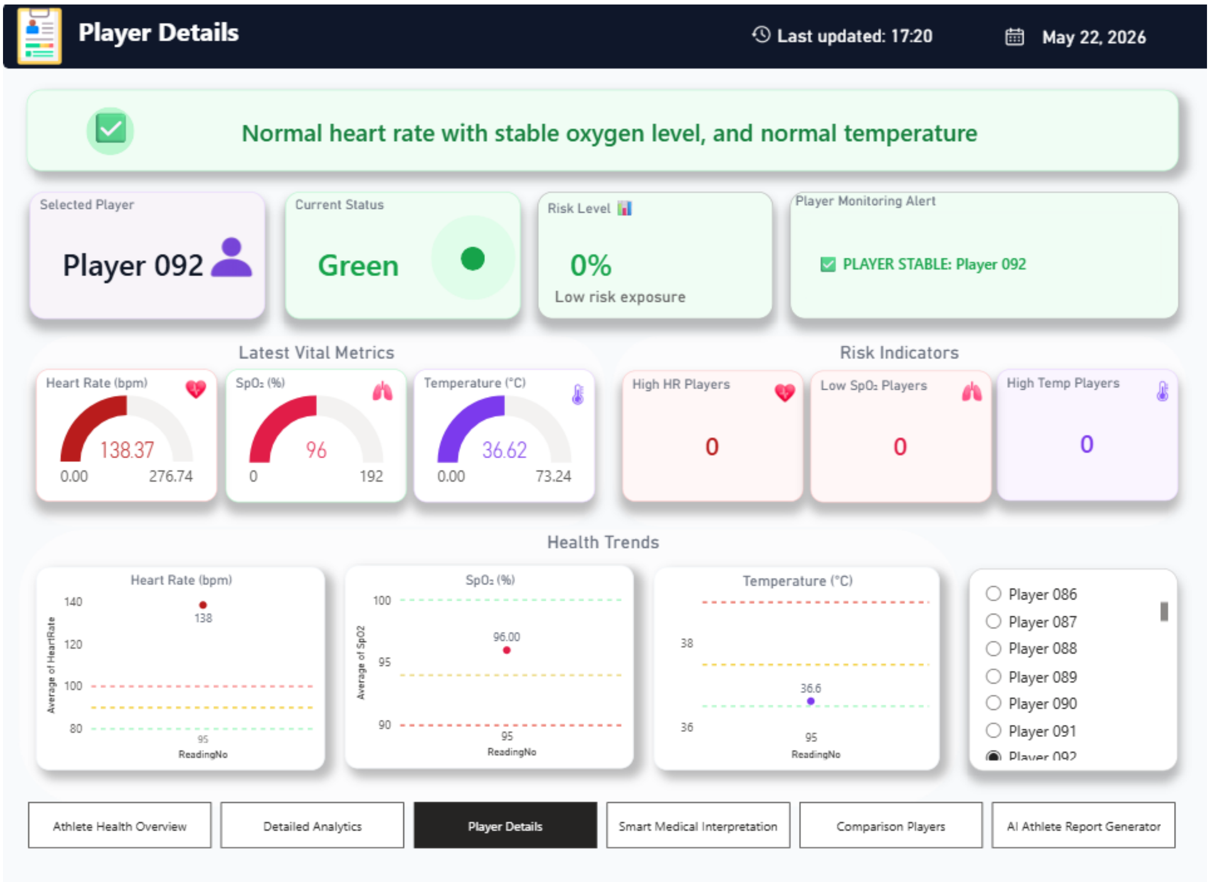

Figure 13Player Details Dashboard Interface

The page includes:
- Current athlete status
- Risk percentage indicators
- Heart Rate, SpO_2, and temperature gauges
- Player monitoring alerts
- Physiological risk indicator cards
- Trend visualizations for each physiological metric
- Interactive athlete selection and filtering components

Gauge visuals were implemented to support rapid interpretation of physiological measurements and monitoring thresholds. Risk-related indicators were visually separated to improve readability and allow users to quickly identify abnormal physiological conditions.

Trend charts were positioned in the lower section of the interface to support longitudinal monitoring and evaluation of athlete physiological behavior over time. The interface dynamically updates when users select different athletes through the interactive filtering panel, enabling real-time exploration of player-specific monitoring data.

Smart Medical Interpretation Interface

The Smart Medical Interpretation page was designed to provide a structured analytical view of athlete physiological conditions and risk-related insights within a clinically oriented dashboard environment. The interface focuses on presenting physiological status summaries, monitoring indicators, and recommendation-oriented analysis in a clear and organized format.

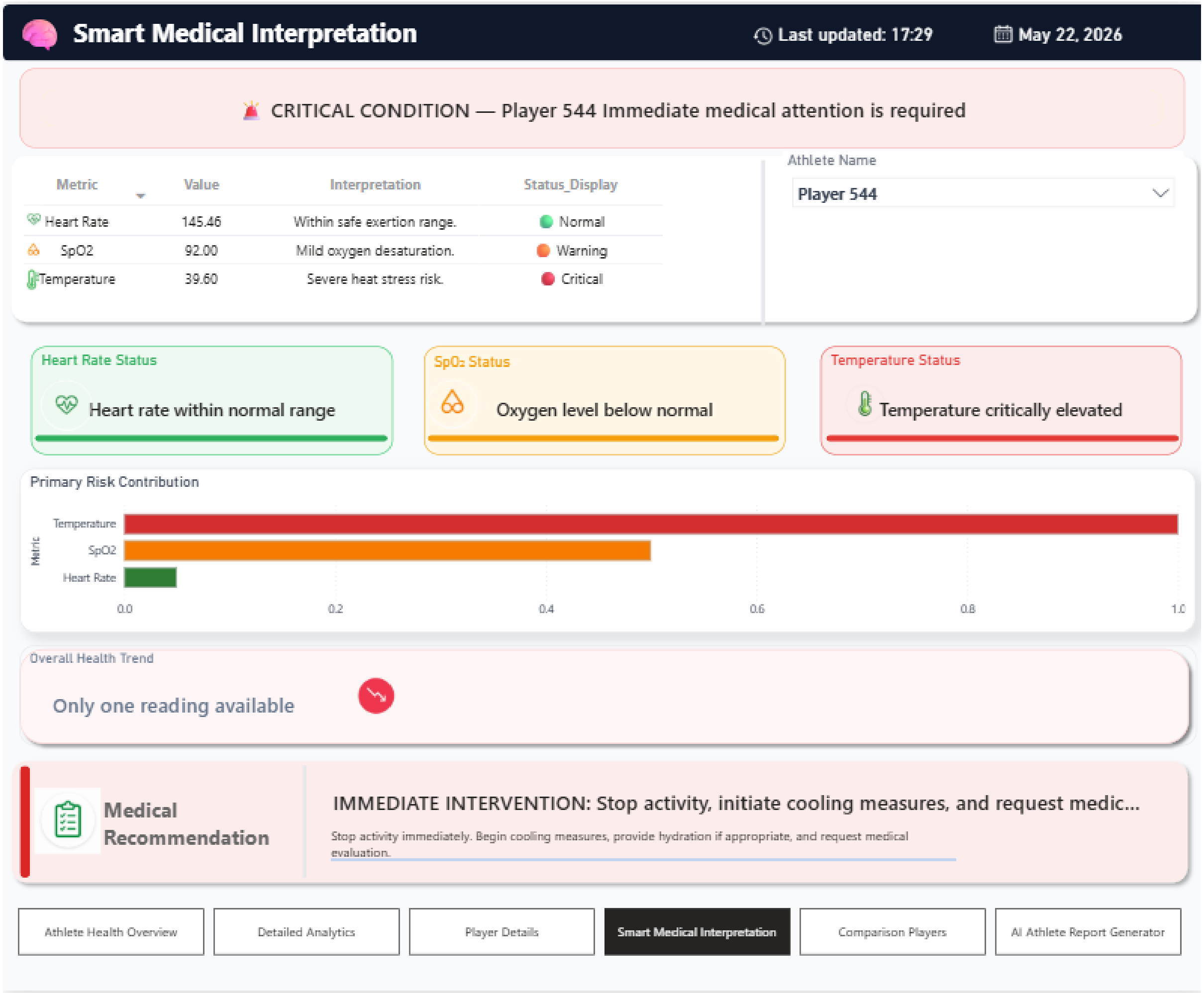

Figure 14Smart Medical Interpretation Dashboard Interface

The page contains:
• Physiological status summaries
• Individual metric assessment
• Health status indicators
• Risk contribution analysis
• Overall health condition evaluation
• Medical recommendation section

The interface organizes physiological findings into structured analytical cards to improve readability and reduce information overload during dashboard interaction. Risk contribution visuals highlight the impact of Heart Rate, SpO_2, and temperature on the athlete's overall condition.

Alert banners and warning indicators were visually emphasized using condition-based color schemes to improve visibility of abnormal physiological measurements and support rapid identification of monitoring priorities.

Comparison Players Interface

The Comparison Players page was designed to support side-by-side physiological comparison between two athletes within an interactive analytical environment. The interface enables users to evaluate differences in physiological measurements, monitoring indicators, and overall risk conditions.

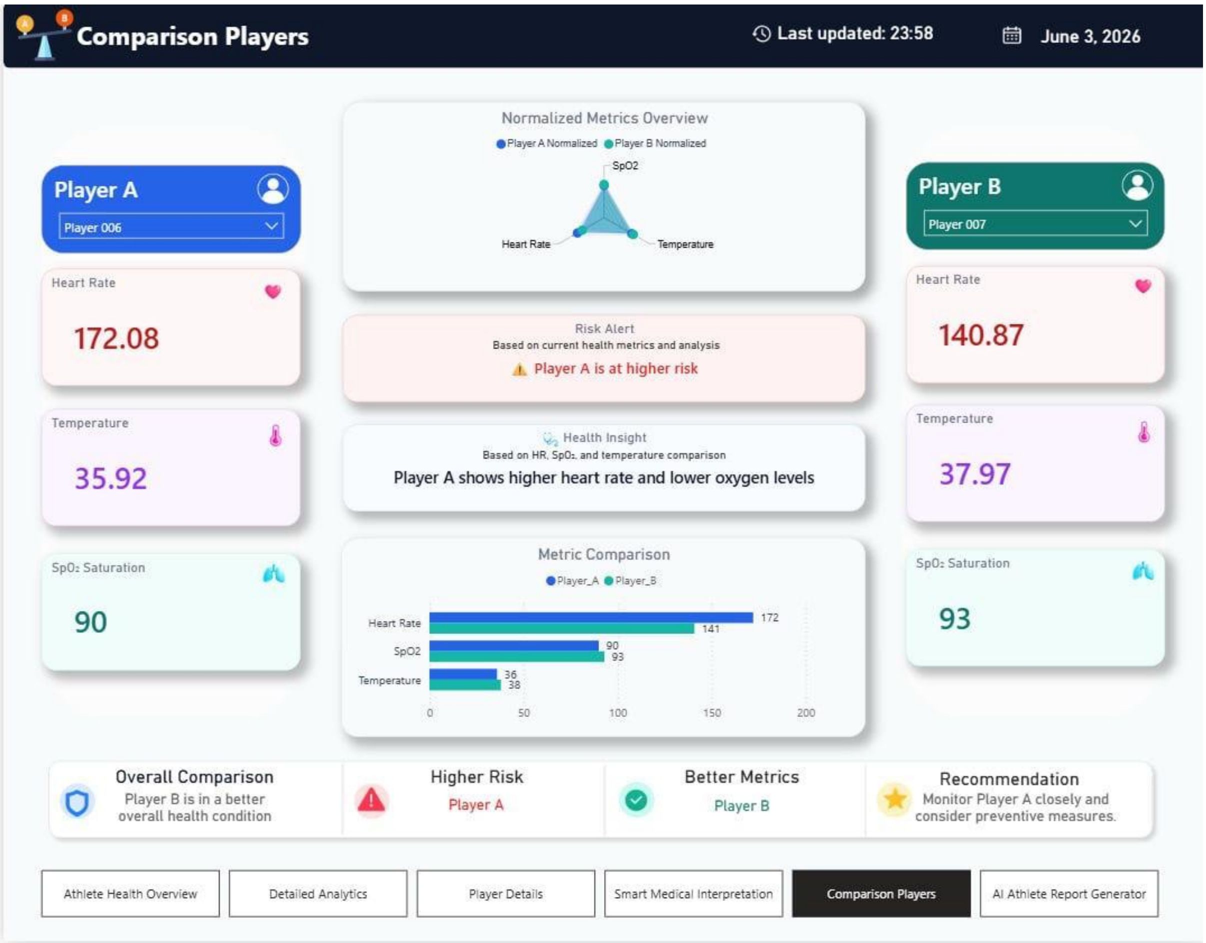

Figure 15Comparison Players Dashboard Interface

The interface includes:
• Athlete comparison cards
• Radar chart visualization
• Metric comparison bar charts
• Risk evaluation section
• Comparative health insights
• Recommendation summaries

The side-by-side layout improves comparative visibility and allows coaches and analysts to quickly identify physiological differences and higher-risk athletes. Normalized visualizations were implemented to support balanced comparison between athletes with different physiological ranges and monitoring conditions.

AI Athlete Report Generator Interface

The AI Athlete Report Generator page was designed to provide structured AI-powered health interpretations and analytical summaries for selected athletes based on physiological monitoring data. This page integrates the Gemini LLM through a Python integration layer to generate automated physiological interpretations and recommendation-oriented analytical reports.

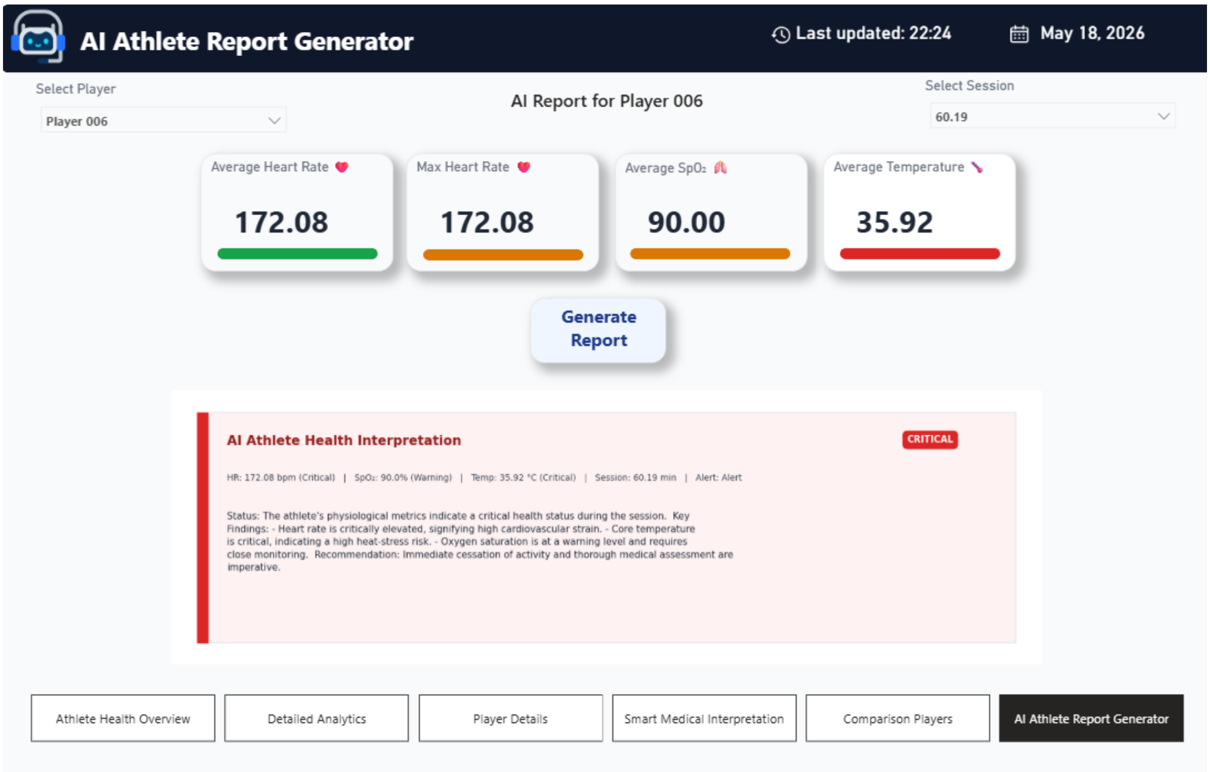

Figure 16AI Athlete Report Generator Dashboard Interface

The interface workflow includes:

1. Athlete and session selection
2. Display of summarized physiological metrics
3. AI report generation trigger
4. Processing physiological data through the Gemini LLM
5. Display of generated interpretation and recommendations

The page presents summarized physiological indicators such as average heart rate, SpO_2 level, and body temperature using KPI cards to provide a quick overview of athlete condition before generating the AI report.

When the user requests report generation, the selected physiological measurements are processed through the Python integration layer, where structured prompts are prepared and sent to the Gemini LLM. The generated response is then formatted and displayed within the dashboard as a structured athlete health interpretation report.

The generated report section was visually separated using a dedicated analytical card layout to improve readability and simulate professional medical-style reporting formats. The report structure includes:

- Athlete physiological status
- AI-generated interpretation
- Key analytical findings
- Risk indicators
- Recommendation-oriented insights

Condition-based status labels and warning indicators were integrated to emphasize abnormal physiological measurements and improve rapid interpretation of athlete conditions.

The interface design intentionally minimizes unnecessary visual complexity in order to maintain user focus on AI-generated analytical interpretations and physiological monitoring insights. The overall layout supports clear report readability while maintaining consistency with the visual structure used across the dashboard environment.

5.12 Usability and User Experience Considerations

Several usability and user experience principles were considered during interface development:

Visual Consistency

Consistent colors, typography, spacing, and dashboard layouts were applied across all pages to improve familiarity and reduce navigation complexity.

Information Hierarchy

Critical physiological alerts and KPI indicators were positioned prominently to support rapid identification of abnormal athlete conditions.

Interactive Navigation

Dashboard tabs and filtering components allow users to transition efficiently between analytical modules and athlete views.

Real-Time Analytical Feedback

Interactive visuals dynamically respond to athlete selection, filters, and session changes, improving responsiveness and analytical exploration.

Clinical-Oriented Design

The dashboard interface was designed to resemble structured medical monitoring systems through the use of status indicators, alert banners, analytical summaries, and physiological gauges.

Reduced Cognitive Load

Visual clutter was minimized by grouping related analytical elements and maintaining balanced spacing between components.

Accessibility and Readability

High-contrast visual elements and structured layouts were used to improve readability during prolonged analytical usage.

Wireframes and Interface Prototyping

Initial dashboard layouts and interface structures were planned using wireframe-style analytical arrangements before final implementation in Power BI. The design process focused on organizing:

- KPI locations
- Analytical chart positioning
- Risk visualization hierarchy
- Navigation structure
- AI interpretation placement
- Comparative analytical workflow

Several iterations of the dashboard layout were evaluated and refined to improve analytical usability, visual balance, and operational clarity.

The final implemented interface reflects the integration of physiological monitoring, AI-based analytics, and interactive visualization within a unified athlete health monitoring environment

5.13 Database / Data Model Design

The data model of the athlete health monitoring platform was designed to support efficient physiological data analysis, dynamic dashboard visualization, and AI-powered analytical processing. The overall design focused on creating a structured and scalable analytical environment capable of supporting interactive reporting, health classification, and intelligent athlete monitoring within Microsoft Power BI.

The project utilized a relational analytical data model in which athlete physiological data was organized into multiple interconnected tables linked through defined relationships. The implemented model was optimized to improve analytical performance, maintain data consistency, and support dynamic filtering and cross-dashboard interaction.

Data Modeling Approach

The data model was developed using a modular analytical structure that separates physiological measurements, calculated metrics, classification logic, and visualization support components into independent but connected entities. This approach improved maintainability, scalability, and analytical flexibility throughout the dashboard environment.

Before modeling, the dataset underwent several preprocessing procedures including:

- Data cleaning
- Duplicate removal
- Data transformation
- Data restructuring
- Validation of physiological values

These preprocessing stages improved data quality and prepared the dataset for relational analytical modeling within Power BI.

Main Data Tables

The implemented data model consisted of several analytical and supporting tables designed to handle different dashboard functionalities and analytical operations.

Table	Purpose
HR_Long	Stores physiological measurements and athlete monitoring data
Athletes_Map	Maintains athlete identification and mapping relationships
Session	Contains session-related information and monitoring records
Measure	Stores calculated analytical measures and KPI calculations
Latest_Player_Status	Maintains latest athlete physiological status and risk indicators
Diagnostic_Metrics	Handles diagnostic interpretation and physiological evaluation
Risk_Contribution	Stores calculated contribution of physiological indicators to overall risk
Player_A_Table / Player_B_Table	Supports multi-player comparison functionality
Status_Sort	Organizes and standardizes athlete status classification
Risk_Status_Table	Maintains risk category logic and classification structure
Risk_Bubbles	Supports risk visualization and bubble chart analytics
Progress_Category	Supports analytical grouping and categorization

Table 6Main Tables in the Power BI Data Model

Relational Data Structure

Relationships were established between tables to support:

- Dynamic filtering
- Athlete-level analysis
- Cross-visual interaction
- Trend monitoring
- Comparative analytics
- AI-based reporting workflows

The relational structure enabled synchronized dashboard behavior across all analytical pages and improved the consistency of calculated results and visual outputs.

Analytical Measures and DAX Calculations

The project extensively utilized DAX (Data Analysis Expressions) to create dynamic analytical measures and calculated fields used throughout the dashboard environment.

These calculations supported:

- Athlete risk percentage calculations
- Dynamic health classification
- KPI generation
- Trend analysis
- Risk contribution analysis
- Multi-player comparison
- Dashboard interaction logic

The analytical measures formed a critical component of the intelligent monitoring framework implemented within the platform.

AI-Based Health Classification Integration

A major component of the data model was the integration of an AI-driven health classification framework based on evidence-based physiological thresholds derived from scientific literature.

The classification logic dynamically evaluated:

- HR values relative to HR Peak
- SpO_2 measurements
- Core body temperature readings

Athlete conditions were automatically categorized into:

- Green (Safe)
- Orange (Warning)
- Red (Critical)

The generated classification outputs were integrated directly into dashboard visuals, KPI cards, risk indicators, analytical charts, and AI-generated reports.

Dashboard Optimization and Scalability

The data model was optimized to improve dashboard responsiveness and visualization performance through:

- Structured relationships
- Organized table hierarchy
- Efficient calculated measures
- Reduced analytical redundancy
- Modular data organization

This optimization enhanced interactive dashboard performance and supported efficient real-time analytical visualization.

AI Interpretation Data Flow

The relational data model also supported the integration of the Gemini LLM through Python scripts. Physiological measurements, calculated indicators, and athlete risk classifications were dynamically transferred to the AI model to generate structured athlete health interpretations and recommendations.

This integration transformed the data model into an intelligent analytical foundation capable of supporting both interactive visualization and AI-powered decision-support functionalities.

Data Model Visualization

Figure illustrates the relational data model and table relationships implemented within the Power BI environment.

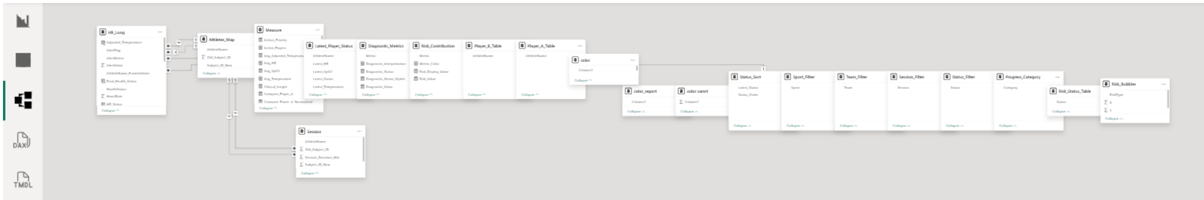

Figure 17Power BI Relational Data Model

The implemented relational data model successfully provided a scalable and efficient analytical foundation for the athlete health monitoring platform, enabling intelligent classification, interactive visualization, dynamic reporting, and AI-powered analytical interpretation within a unified system environment.

5.14 Algorithm / Process Design

The algorithm and process design of the VYTAL athlete health monitoring platform focused on developing an intelligent analytical workflow capable of processing athlete physiological data, performing evidence-based health classification, generating interactive analytical insights, and producing AI-assisted interpretation outputs. The overall process design combined data preprocessing, analytical modeling, threshold-based classification, dashboard interaction, Python execution, and Gemini-powered interpretation within a unified monitoring environment.

The system workflow begins with importing athlete physiological data obtained from the Kaggle dataset into Microsoft Power BI. The imported dataset contains physiological indicators and monitoring variables including Heart Rate, SpO_2, body temperature, session duration, athlete identifiers, and alert indicators.

After data import, preprocessing and transformation procedures are executed using Power Query and Python. The preprocessing stage includes:

- Data cleaning
- Duplicate removal
- Missing value handling
- Physiological value validation
- Data restructuring
- Attribute standardization
- Session-based organization

These procedures ensure that the dataset is suitable for analytical calculations, dashboard visualization, and AI-assisted interpretation workflows.

Following preprocessing, a relational analytical data model is constructed inside Power BI. Relationships between analytical tables are configured to support synchronized filtering, cross-dashboard interaction, player-level analysis, and dynamic visualization updates. DAX calculations and calculated fields are then implemented to generate:

- KPI indicators
- Risk percentages
- Dynamic athlete status labels
- Trend calculations
- Comparative analytics
- Diagnostic summaries
- Risk contribution metrics

A major component of the system process design is the AI-based health classification algorithm. The implemented classification logic evaluates athlete physiological measurements using evidence-based threshold rules derived from scientific medical literature. The classification algorithm analyzes:

- HR values relative to HR Peak
- Oxygen saturation
- Core body temperature

Based on the evaluated physiological values, the system dynamically classifies athlete conditions into:

- Green (Normal)
- Orange (Warning)
- Red (Critical)

The classification process is implemented through DAX conditional logic and analytical threshold evaluation rules inside Power BI. Once the classification is calculated, the generated outputs automatically update dashboard KPI cards, risk indicators, charts, trend visuals, and monitoring summaries.

The analytical workflow then proceeds to the dashboard visualization stage. Interactive dashboard pages retrieve the processed analytical outputs and dynamically present them through:

- KPI cards
- Donut charts
- Trend graphs
- Comparative charts
- Dynamic tables
- Gauge indicators
- Risk distribution visuals
- Heatmap-style analytical views

Interactive slicers and filters allow users to select athletes, sessions, and analytical categories, triggering synchronized updates across all dashboard components.

One of the most advanced processes implemented in the system involves the Python execution and AI interpretation workflow. When a user selects an athlete and requests an AI interpretation, Python scripts embedded inside Power BI execute the following process:

1. Retrieve selected athlete physiological measurements
2. Validate physiological inputs
3. Apply threshold-based analytical checks
4. Generate structured AI prompts
5. Send prompts to the Gemini 2.5 Flash LLM through the Google Generative AI API
6. Receive AI-generated interpretation responses
7. Format and render analytical interpretation outputs inside the dashboard

The generated prompts include:

- Physiological measurements
- Risk classifications
- Alert indicators
- Analytical context
- Formatting instructions
- Recommendation requirements
- Safety constraints

The Gemini LLM generates concise analytical reports containing:

- Athlete condition summaries
- Key physiological findings
- Risk-oriented observations
- Monitoring insights
- Recommendation-based outputs

The generated interpretation is then formatted dynamically using Python and Matplotlib rendering techniques to produce structured AI-powered athlete report cards displayed directly inside the dashboard interface.

Exception-handling algorithms were also implemented to improve system reliability. If API communication fails or invalid responses are returned, fallback analytical summaries are automatically generated using the existing dashboard calculations and physiological classifications.

The complete system process follows a structured analytical pipeline consisting of:

1. Data acquisition
2. Data preprocessing and validation
3. Relational analytical modeling
4. DAX-based analytical calculations
5. AI-based physiological classification
6. Interactive dashboard visualization
7. Python execution workflow
8. Gemini AI interpretation generation
9. Dynamic report rendering
10. Real-time analytical interaction and monitoring

This algorithm and process design enabled the successful development of an intelligent athlete health monitoring platform capable of integrating physiological analytics, interactive visualization, automated classification, and AI-assisted interpretation within a unified analytical decision-support environment.

Chapter 6

Implementation / Experimentation

6.1 Implementation Plan

The implementation phase focused on developing an intelligent athlete health monitoring platform capable of analyzing physiological data, classifying athlete health conditions, generating AI-powered interpretations, and presenting interactive analytical visualizations through Microsoft Power BI and Python integration. The system was implemented through multiple structured development phases to ensure analytical accuracy, modular integration, interactive responsiveness, and AI-assisted analytical functionality.

The implementation process combined business intelligence technologies, physiological analytics, artificial intelligence integration, and interactive visualization techniques to transform raw athlete physiological measurements into meaningful health insights and decision-support outputs.

Development Environment and Technologies

The system was implemented using a combination of analytical, visualization, and AI technologies. Microsoft Power BI served as the primary development environment for dashboard construction, relational data modeling, DAX calculations, and interactive visualization. Python was integrated within Power BI to support advanced analytical processing, AI workflow execution, prompt generation, and communication with the Gemini Large Language Model (LLM).

The main technologies and tools used during implementation included:

- Microsoft Power BI
- Power Query
- DAX (Data Analysis Expressions)
- Python
- Gemini 2.5 Flash LLM
- Google Generative AI API
- Pandas
- Matplotlib
- Kaggle Athlete Dataset

The implementation environment enabled seamless integration between data preprocessing, analytical modeling, AI-based health classification, dashboard interaction, and automated AI-powered report generation.

Dataset and Data Preparation Phase

The implementation process began with importing athlete physiological data obtained from a Kaggle dataset containing multiple monitoring variables and physiological indicators related to athlete health conditions and training sessions.

The dataset included several important physiological attributes, including:

• Heart Rate
• Oxygen Saturation
• Body Temperature
• Session Duration
• Alert Indicators
• Athlete Identifiers
• Session Information

After data collection, a dedicated preprocessing phase was conducted using Power Query and Python to improve data quality and prepare the dataset for analytical processing.

The preprocessing tasks included:

• Data cleaning
• Duplicate removal
• Missing value handling
• Data transformation
• Data restructuring
• Formatting physiological variables
• Standardizing attribute names
• Creating structured analytical tables

Additional preprocessing procedures were implemented using Python and Pandas to validate physiological values, convert numerical attributes into structured formats, and prepare selected athlete measurements for AI interpretation workflows.

This preprocessing phase ensured that the dataset became suitable for dynamic analysis, AI-based classification, dashboard visualization, and automated interpretation generation.

Data Modeling and Analytical Structure Development

After preprocessing, a relational analytical data model was constructed inside Microsoft Power BI. The data model was designed to support synchronized filtering, dynamic calculations, interactive analytics, and AI-powered reporting workflows.

Several analytical tables and entities were implemented, including:

• Athletes_Map
• HR_Long
• Session
• Diagnostic_Metrics
• Risk_Contribution
• Risk_Status_Table
• Latest_Player_Status

Relationships between tables were established to support accurate analytical interactions across dashboard pages and ensure consistency between filters, visualizations, and calculated measures.

Additionally, multiple DAX measures and calculated fields were implemented to generate:

• KPI indicators
• Risk percentages
• Dynamic athlete status labels
• Comparative analytics
• Trend calculations
• Diagnostic summaries
• Risk contribution values
• Real-time monitoring indicators

The analytical structure enabled efficient aggregation and processing of physiological measurements while supporting dynamic dashboard responsiveness and real-time analytical updates.

AI-Based Health Classification Implementation

One of the most important implementation phases involved developing the AI-based health classification engine. The classification framework was implemented using evidence-based physiological threshold rules derived from published scientific and medical literature.

Athlete physiological conditions were dynamically classified into:

• Green (Normal)
• Orange (Warning)
• Red (Critical)

The classification process analyzed multiple physiological indicators, including:

• Heart Rate ranges
• HR peak percentage
• Oxygen saturation ranges
• Core body temperature values

The classification logic was implemented using DAX calculations and analytical conditional rules inside Power BI. The implemented logic automatically evaluated athlete physiological conditions and generated dynamic risk classifications that updated dashboard visuals, KPI cards, analytical summaries, and monitoring indicators in real time.

This implementation transformed raw physiological measurements into interpretable analytical risk indicators capable of supporting proactive athlete monitoring and health-risk identification.

Dashboard Development and Visualization Phase

The dashboard development phase focused on constructing interactive analytical interfaces capable of supporting athlete monitoring, physiological analysis, and AI-assisted interpretation.

Multiple analytical dashboard pages were implemented, including:

• Athlete Health Overview
• Detailed Analytics
• Player Details
• Player Comparison
• Smart Medical Interpretation
• AI Athlete Health Report Generator

The visualization layer included several analytical visual components, such as:

• KPI cards
• Trend analysis charts
• Donut charts
• Comparative visualizations
• Dynamic tables
• Risk distribution visuals
• Gauge indicators
• Interactive slicers and filters
• Diagnostic summary panels

The dashboard interface was designed using a structured analytical layout intended to improve readability, navigation efficiency, visual consistency, and situational awareness during athlete monitoring workflows.

The implemented dashboard structure followed a progressive analytical flow that allowed users to move naturally from high-level monitoring indicators toward detailed player-level analysis and AI-generated interpretation outputs.

Python and Gemini LLM Integration

Python integration represented one of the most advanced implementation stages of the project. Python scripts were integrated directly within Power BI to process selected athlete physiological data and communicate with the Gemini 2.5 Flash Large Language Model through the Google Generative AI API.

The implementation workflow began by retrieving selected athlete physiological measurements from the Power BI dataset. Python preprocessing procedures were then applied using Pandas to clean and structure the selected data before generating AI prompts.

The AI workflow included:

• Reading selected athlete measurements
• Validating physiological values
• Applying threshold-based classifications
• Building structured AI prompts
• Sending prompts to Gemini LLM
• Processing AI-generated responses
• Formatting interpretation outputs
• Rendering AI-generated report cards inside Power BI

The implemented prompt-engineering workflow incorporated:

• Physiological measurements
• Risk classifications
• Alert indicators
• Threshold-based interpretations
• Response formatting instructions
• Safety and consistency constraints

The prompts were designed to ensure that generated responses remained concise, structured, medically readable, and consistent with the analytical classifications already calculated inside the dashboard.

The generated AI reports included:

• Athlete health status
• Key physiological findings
• Risk-oriented interpretation
• Monitoring insights
• Recommendation-based summaries

Matplotlib visualization rendering was additionally integrated to dynamically generate structured AI interpretation cards with condition-based colors, status indicators, and alert-oriented formatting directly inside the dashboard environment.

Exception-handling mechanisms were also implemented to manage API failures, unavailable responses, or invalid outputs. In these situations, fallback analytical summaries were generated automatically using existing physiological classifications and calculated dashboard measures.

This implementation transformed the platform from a traditional analytical dashboard into an intelligent AI-assisted athlete monitoring environment capable of generating automated physiological interpretations and contextual analytical insights.

Testing and Refinement Phase

After integrating all system components, multiple testing and refinement procedures were conducted to ensure analytical consistency, dashboard responsiveness, AI workflow reliability, and visualization stability.

The testing process included:

• Validation of DAX calculations
• Verification of classification outputs
• Dashboard interaction testing
• Filter synchronization testing
• AI report generation testing
• Data relationship validation
• Visualization consistency review
• Python workflow testing
• Gemini response validation
• Risk classification scenario testing

Different physiological scenarios representing Green, Orange, and Red athlete conditions were tested to ensure that dashboard visuals, classification outputs, and AI-generated interpretations responded correctly to changing physiological values.

Several refinements and optimizations were implemented throughout development to improve dashboard performance, visualization clarity, analytical accuracy, and AI response consistency.

The final implemented system successfully integrated business intelligence technologies, physiological analytics, artificial intelligence, and interactive visualization into a unified athlete health monitoring platform capable of supporting intelligent sports-health analysis, risk monitoring, and AI-assisted decision-making.

6.2 Module / Process Development

The athlete health monitoring platform was implemented as a set of interconnected analytical and AI-powered modules. Each module was designed to perform a specific function within the system while supporting integration with the remaining components. This modular structure improved scalability, maintainability, analytical flexibility, and system organization.

The core modules implemented in the system included:

- Data Preprocessing Module
- Data Modeling and Analytical Processing Module
- AI-Based Health Classification Module
- Interactive Visualization Module
- Diagnostic and Risk Analysis Module
- Python Integration Module
- AI-Powered Report Generation Module

Data Preprocessing Module

The data preprocessing module is responsible for preparing the raw athlete dataset for analysis, visualization, and AI-based interpretation. This module was implemented using Power Query within Microsoft Power BI, supported by Python-based processing where needed.

The preprocessing workflow included removing duplicate records, handling inconsistent values, formatting physiological variables, restructuring data tables, standardizing attribute names, filtering unnecessary fields, and organizing session-based measurements. The processed physiological indicators included Heart Rate, SpO_2, body temperature, session duration, and alert flags.

This module ensured that the dataset was clean, structured, and suitable for accurate analytical calculations, dashboard visualization, and AI-powered reporting.

Data Modeling and Analytical Processing Module

The analytical processing module was implemented inside Microsoft Power BI using a relational data model. The model was designed to support dynamic filtering, interactive analysis, synchronized dashboard behavior, and responsive visual updates.

Several structured analytical tables were developed, including Athletes_Map, HR_Long, Diagnostic_Metrics, Latest_Player_Status, Risk_Contribution, Risk_Status_Table, Session. Relationships between these tables were configured to ensure accurate filtering and consistent interaction across dashboard pages.

DAX measures and calculated fields were implemented to generate dynamic KPI calculations, risk percentages, athlete status indicators, trend analytics, comparative metrics, diagnostic summaries, visual status labels, and risk contribution values. This module formed the analytical engine of the platform and enabled real-time-style dashboard updates during user interaction.

AI-Based Health Classification Module

The AI-based health classification module was implemented to dynamically evaluate athlete physiological conditions using evidence-based threshold rules derived from scientific medical literature. This module classified athlete conditions into three categories: Green (Normal), Orange (Warning), and Red (Critical).

The classification engine evaluated Heart Rate, HR peak percentage, SpO_2 level, and core body temperature. The classification logic was implemented through DAX calculations and conditional analytical rules inside Power BI. Once calculated, the classification results were reflected across KPI cards, risk alerts, tables, charts, and summary indicators.

This module transformed raw physiological measurements into interpretable health intelligence, enabling automated athlete status identification, risk visualization, and proactive monitoring.

Interactive Visualization Module

The interactive visualization module was responsible for presenting analytical results through a structured and user-friendly dashboard interface. Several dashboard pages were developed, including Athlete Health Overview, Detailed Analytics, Player Details, Player Comparison, Smart Medical Interpretation, and AI Athlete Health Report Generator.

The visualization layer included KPI cards, donut charts, trend graphs, comparative charts, interactive tables, gauge visuals, heatmap-style visuals, risk distribution indicators, and diagnostic summary panels. Interactive slicers and filters allowed users to select athletes, filter sessions, compare players, explore physiological trends, and analyze risk conditions dynamically.

This module emphasized clarity, responsiveness, consistent visual hierarchy, and reduced cognitive load during athlete monitoring.

Diagnostic and Risk Analysis Module

The diagnostic and risk analysis module was implemented to provide deeper insight into athlete physiological conditions and the factors contributing to risk status. This module analyzed physiological readings and generated diagnostic metrics, risk contribution values, physiological summaries, comparative indicators, and trend-based monitoring insights.

The system identified which physiological indicator contributed most significantly to an athlete's risk condition. This improved explainability by showing whether risk was mainly associated with elevated Heart Rate, reduced SpO_2, abnormal temperature, or combined physiological stress.

This module strengthened the interpretability of the classification outputs and supported clearer decision-making for coaches and medical teams.

Python Integration Module

The Python integration module enabled advanced processing and communication between Power BI and the Gemini LLM. Python scripts were embedded within Power BI to read selected athlete data, validate physiological measurements, prepare structured prompts, send requests to Gemini, process AI-generated responses, and format interpretation outputs.

The implementation used Python libraries such as Pandas, Matplotlib, Textwrap, Regular Expressions, and the Google Generative AI SDK. Python acted as the intermediary layer between dashboard analytics and AI-generated interpretation.

The workflow also included exception handling to manage API errors, unavailable responses, or invalid outputs. When AI generation was unavailable, the system produced fallback analytical summaries based on the existing physiological classifications and dashboard calculations.

AI-Powered Report Generation Module

The AI-powered report generation module was implemented to produce structured athlete health interpretation reports based on physiological measurements, alert indicators, risk classification results, and diagnostic outputs.

The Gemini LLM was integrated into the platform through a structured prompt-engineering workflow designed to generate consistent, context-aware, and analytically relevant athlete health interpretations based on physiological measurements and calculated risk classifications. The prompt included selected physiological values, calculated classifications, alert status, threshold-based context, formatting instructions, and response constraints. This ensured that generated reports remained concise, consistent, and aligned with the system's classification logic.

The generated reports included athlete condition summaries, key physiological findings, risk-oriented interpretations, monitoring insights, and recommendation-based outputs. Matplotlib was used to render the generated interpretation inside Power BI as a structured report card with status labels, condition-based colors, and alert-oriented formatting.

Overall Module Integration

All modules were connected within a unified analytical workflow. The system begins with preprocessing and structuring the dataset, then applies relational modeling and DAX calculations, performs AI-based classification, updates interactive visualizations, and generates AI-powered interpretation reports through Python and Gemini integration.

This modular implementation produced a scalable and intelligent athlete monitoring platform capable of supporting physiological analysis, automated risk classification, interactive dashboard exploration, and AI-assisted decision support.

6.3 Integration of Components

The system components were integrated progressively to ensure that each module functioned correctly before being connected to the complete athlete health monitoring platform. The integration process focused on combining data preprocessing, relational modeling, analytical calculations, AI-based classification, dashboard visualization, Python execution, and Gemini-powered interpretation into one unified intelligent analytical environment.

The integration workflow began by importing the cleaned athlete physiological dataset into Microsoft Power BI. After importing the dataset, structured analytical tables were created and connected through relational mappings to support synchronized filtering, session-based analysis, player-level monitoring, and cross-dashboard interaction.

Several analytical tables were integrated into the relational model, including:

- HR_Long
- Athletes_Map
- Session
- Latest_Player_Status
- Diagnostic_Metrics
- Risk_Contribution
- Risk_Status_Table

Relationships between these tables were configured and validated to ensure accurate interaction between dashboard visuals, filters, analytical measures, and AI-generated outputs.

After establishing the relational data model, DAX measures and calculated fields were integrated into the dashboard logic. These measures were responsible for generating:

- KPI indicators
- Risk percentages
- Latest athlete readings
- Diagnostic summaries
- Comparative metrics
- Trend calculations
- Dynamic status labels
- Risk contribution values

The implemented DAX calculations were tested across multiple dashboard pages to ensure that analytical outputs updated correctly whenever users selected different athletes, sessions, or risk categories.

Following the analytical model integration, the AI-based health classification component was connected to the dashboard workflow. Physiological indicators including Heart Rate, SpO_2, and body temperature were evaluated using evidence-based threshold rules implemented through DAX and conditional analytical logic.

The classification outputs were integrated directly with:

- KPI cards
- Risk indicators
- Trend charts
- Comparative visuals
- Diagnostic summaries
- Risk contribution panels
- Dashboard status banners

This integration ensured that the same classification logic was reflected consistently across all analytical components and dashboard pages.

After validating the analytical calculations and classification outputs, the interactive visualization layer was integrated with the shared analytical model. Dashboard pages such as Athlete Health Overview, Detailed Analytics, Player Details, Player Comparison, Smart Medical Interpretation, and AI Athlete Report Generator were connected to the centralized dataset and analytical calculations.

Interactive slicers and filters were then tested to ensure synchronized behavior across:

- Charts
- Tables
- KPI cards
- Risk indicators
- AI-generated outputs
- Trend visualizations

The testing confirmed that user selections dynamically updated all related dashboard components without calculation conflicts or visualization inconsistencies.

The Python integration layer was then connected to the selected athlete data inside Power BI. Python scripts were embedded directly within the dashboard environment to retrieve selected physiological measurements, validate analytical inputs, generate structured prompts, communicate with Gemini LLM, and process AI-generated responses.

The implemented Python workflow included:

- Reading selected athlete physiological values
- Validating and formatting measurements
- Applying threshold-based classifications
- Constructing structured AI prompts
- Sending requests through the Google Generative AI API
- Processing generated responses
- Formatting analytical interpretation outputs
- Rendering AI interpretation cards using Matplotlib

This integration created a direct workflow between dashboard interaction and automated AI-powered analytical interpretation.

The Gemini 2.5 Flash LLM was integrated as the final intelligent reporting layer within the system architecture. The model received selected physiological values including Heart Rate, SpO_2, adjusted body temperature, session duration, alert status, and calculated classification outputs.

A structured prompt-engineering workflow was implemented to control the consistency, readability, and analytical reliability of generated AI responses. The prompts included:

- Physiological measurements
- Calculated classifications
- Alert indicators
- Risk-oriented analytical context
- Formatting instructions
- Response constraints

The generated AI interpretations were designed to produce:

- Athlete condition summaries
- Key physiological findings
- Risk-oriented observations
- Monitoring insights
- Recommendation-based outputs

The generated response was then formatted dynamically and displayed within the dashboard as an AI-powered athlete interpretation card.

Several testing and validation procedures were conducted throughout the integration process to ensure system reliability, analytical consistency, and AI workflow stability. The testing procedures included:

- Validation of table relationships
- Verification of DAX calculations
- Dashboard interaction testing
- Filter synchronization testing
- Risk classification validation
- Visualization consistency review
- Python workflow testing
- Gemini response validation
- AI report generation testing
- Dashboard responsiveness testing

Different athlete scenarios representing Green, Orange, and Red physiological conditions were tested to confirm that the dashboard, classification engine, analytical measures, and AI-generated reports responded correctly to changing physiological measurements.

Exception-handling mechanisms were additionally implemented to manage API failures, unavailable AI responses, or invalid generated outputs. In these situations, fallback analytical summaries were generated automatically using existing physiological classifications and calculated dashboard measures.

The final integration process successfully connected data preprocessing, relational modeling, analytical calculations, AI-based classification, interactive visualization, Python execution, and Gemini-powered interpretation into one complete athlete health monitoring platform capable of supporting intelligent physiological monitoring, automated analytical interpretation, and AI-assisted decision-making.

6.4 User Interface / Backend Development

The development of the VYTAL athlete health monitoring platform focused on integrating an interactive analytical user interface with a dynamic backend analytical workflow capable of processing physiological data, generating real-time analytical insights, and supporting AI-powered interpretation.

The implementation combined front-end dashboard visualization techniques with backend analytical processing, classification logic, Python execution workflows, and AI integration to create a unified intelligent monitoring environment.

User Interface Development

The user interface was developed using Microsoft Power BI and designed to provide a structured, visually intuitive, and analytically efficient monitoring environment for coaches, medical teams, analysts, and athletes.

The dashboard interface followed a modular analytical design approach where each page was developed to support a specific monitoring or analytical objective. The implemented dashboard pages included:

- Athlete Health Overview
- Detailed Analytics
- Player Details
- Player Comparison
- Smart Medical Interpretation
- AI Athlete Health Report Generator

The interface design emphasized:

- Visual clarity
- Analytical readability
- Structured information hierarchy
- Interactive responsiveness
- Reduced cognitive overload
- Rapid risk identification

Several interactive visual components were implemented throughout the dashboard, including:

- KPI cards
- Donut charts
- Trend analysis graphs
- Comparative visualizations
- Dynamic tables
- Gauge indicators
- Risk distribution visuals
- Heatmap-style analytical visuals
- Interactive slicers and filters

The dashboard layout was designed to support progressive analytical navigation, allowing users to move naturally from high-level athlete monitoring indicators toward detailed physiological analysis and AI-generated interpretation outputs.

Condition-based color mapping was implemented consistently across the interface to improve rapid interpretation of athlete health conditions:

- Green → Normal condition
- Orange → Warning condition
- Red → Critical condition

Dynamic filters and slicers were integrated to allow users to:

- Select individual athletes
- Filter monitoring sessions
- Compare multiple players
- Explore physiological trends
- Analyze risk distributions dynamically

The interface was designed using a card-based analytical layout with structured spacing, consistent typography, and synchronized visual styling to maintain usability and analytical consistency across all dashboard pages.

Special attention was given to the AI interpretation interfaces to ensure that generated analytical outputs were displayed in a readable and professionally structured format resembling concise medical-style monitoring reports.

Backend Analytical Processing Development

Although the system was implemented primarily as an analytical dashboard environment rather than a traditional web-based application, several backend analytical workflows and processing layers were developed to support data execution, classification logic, AI integration, and automated interpretation generation.

The backend analytical workflow began with preprocessing the athlete physiological dataset using Power Query and Python. Raw physiological measurements were cleaned, validated, transformed, and structured into analytical tables suitable for dashboard processing.

The preprocessing workflow included:

- Data cleaning
- Duplicate removal
- Missing value handling
- Attribute formatting
- Data restructuring
- Session organization
- Physiological value validation

After preprocessing, a relational analytical model was implemented inside Microsoft Power BI. Relationships between analytical tables were configured to support synchronized filtering, cross-dashboard interaction, and dynamic calculation updates.

Several DAX measures and calculated fields were implemented as part of the backend analytical logic to generate:

- KPI calculations
- Athlete status indicators
- Risk percentages
- Trend analytics
- Comparative metrics
- Diagnostic summaries
- Risk contribution calculations
- Dynamic monitoring indicators

The backend processing layer also included the implementation of the AI-based health classification engine. This analytical engine evaluated physiological indicators including Heart Rate, SpO_2, HR peak percentage, and body temperature using evidence-based threshold rules derived from scientific literature.

The implemented classification logic dynamically categorized athlete conditions into:

- Green (Normal)
- Orange (Warning)
- Red (Critical)

The classification outputs were automatically synchronized with dashboard visuals, KPI cards, trend charts, analytical summaries, and alert indicators.

Python Execution and AI Workflow

One of the most advanced backend implementation components involved integrating Python execution workflows within Power BI to support AI-assisted analytical interpretation.

Python scripts were embedded directly inside the dashboard environment to perform:

- Data extraction from selected dashboard filters
- Physiological value validation
- Threshold-based classification processing
- Prompt generation
- Communication with Gemini LLM
- AI response processing
- Automated interpretation rendering

The implementation utilized several Python libraries, including:

- Pandas
- Matplotlib
- Textwrap
- Regular Expressions (re)
- Google Generative AI SDK

The Python workflow began by retrieving selected athlete physiological measurements directly from the Power BI dataset. The selected data included:

- Heart Rate
- SpO_2
- Adjusted body temperature
- Session duration
- Alert status

The retrieved measurements were processed and validated using Pandas before being analyzed using threshold-based analytical logic.

The backend workflow then generated structured prompts containing:

- Physiological measurements
- Calculated classifications
- Risk-oriented analytical context
- Formatting instructions
- Safety constraints
- Recommendation requirements

These prompts were transmitted to the Gemini 2.5 Flash Large Language Model through the Google Generative AI API.

The AI-generated response was then processed, formatted, and rendered dynamically inside the dashboard interface using Matplotlib visualization rendering techniques. The generated interpretation cards included:

- Athlete status indicators
- Key physiological findings
- Risk-oriented observations
- Monitoring insights
- Recommendation-based summaries

Condition-based color mapping and structured report formatting were implemented to improve readability and visual interpretability.

Data Processing and Execution Workflow

The overall backend execution workflow followed a structured analytical pipeline consisting of:

1. Importing athlete physiological data
2. Cleaning and preprocessing the dataset
3. Building relational analytical tables
4. Implementing DAX calculations and analytical measures
5. Applying threshold-based health classification logic
6. Updating dashboard visualizations dynamically
7. Retrieving selected athlete measurements
8. Executing Python analytical workflows
9. Generating structured AI prompts
10. Processing Gemini-generated analytical interpretations
11. Rendering AI-powered interpretation reports inside the dashboard

This workflow enabled seamless interaction between dashboard analytics, physiological classification, AI-assisted interpretation, and interactive visualization.

System Responsiveness and Execution Validation

Several validation and refinement procedures were implemented during backend development to ensure reliable execution and analytical consistency.

The implemented testing procedures included:

- DAX calculation validation
- Dashboard interaction testing
- Filter synchronization testing
- Risk classification verification
- Python workflow validation
- AI response consistency testing
- Visualization responsiveness testing
- Analytical output validation

Different physiological scenarios representing normal, warning, and critical athlete conditions were tested to ensure that dashboard analytics and AI-generated outputs responded correctly to changing physiological measurements.

Exception-handling mechanisms were additionally implemented within the Python workflow to manage invalid data, unavailable AI responses, API limitations, and formatting inconsistencies. In these situations, fallback analytical summaries were generated automatically using existing dashboard calculations and classification outputs.

The final implementation successfully combined interactive user interface development, backend analytical processing, AI integration, and automated interpretation workflows into a unified intelligent athlete health monitoring environment capable of supporting real-time analytical monitoring and AI-assisted decision-making.

6.5 Deployment or Execution

The final VYTAL athlete health monitoring platform was deployed using Microsoft Power BI Service to provide a cloud-based interactive analytical environment accessible through web browsers. The deployment phase focused on enabling centralized dashboard access, dynamic interaction, analytical responsiveness, and AI-assisted monitoring capabilities within a unified execution environment.

After completing dashboard development and component integration inside Microsoft Power BI Desktop, the final dashboard file was published to the Power BI Service platform. This deployment allowed the system to operate as an online analytical dashboard capable of supporting interactive athlete monitoring and real-time-style analytical visualization.

The deployed dashboard can be accessed through the following Power BI Service link:

[Power BI Dashboard Link Here]

The deployed environment enabled users to:

- Access the dashboard remotely through web browsers
- Navigate between analytical dashboard pages
- Interact with charts, KPI cards, and monitoring visuals
- Apply filters and slicers dynamically
- Explore athlete physiological trends
- Review AI-generated analytical interpretations
- Monitor athlete risk conditions interactively

The deployed dashboard environment included all implemented analytical pages, including:

- Athlete Health Overview
- Detailed Analytics
- Player Details
- Player Comparison
- Smart Medical Interpretation
- AI Athlete Health Report Generator

The deployment workflow also preserved the relational analytical model, DAX calculations, classification logic, dashboard interactions, and synchronized filtering behavior developed during implementation.

The execution workflow inside the deployed environment followed several analytical stages:

1. Loading athlete physiological data
2. Processing analytical calculations and DAX measures
3. Applying threshold-based health classification logic
4. Updating dashboard visualizations dynamically
5. Processing selected athlete measurements
6. Executing Python analytical workflows
7. Generating AI-powered interpretation outputs
8. Displaying structured analytical reports inside the dashboard

The deployed system maintained interactive responsiveness between dashboard visuals, filters, analytical calculations, and AI-generated interpretation outputs. User interactions automatically triggered synchronized updates across KPI indicators, charts, tables, risk visualizations, and monitoring summaries.

The AI-assisted interpretation workflow was executed through embedded Python integration connected to the Gemini 2.5 Flash Large Language Model using the Google Generative AI API. Selected athlete physiological measurements were processed dynamically to generate structured analytical interpretations and recommendation-oriented outputs directly within the dashboard environment.

Several validation procedures were performed after deployment to verify system stability and execution consistency. These procedures included:

- Dashboard accessibility testing
- Filter interaction validation
- DAX calculation verification
- Visualization responsiveness testing
- Cross-page synchronization testing
- Risk classification consistency review

Different physiological scenarios representing normal, warning, and critical athlete conditions were tested to confirm reliable execution behavior within the deployed analytical environment.

The final deployed system successfully provided a cloud-based intelligent athlete health monitoring platform capable of combining physiological analytics, interactive visualization, automated classification, and AI-assisted interpretation within one integrated analytical environment.

Chapter 7

Results and Evaluation

7.1 Results Presentation

This section presents the results obtained from implementing the VYTAL Athlete Health Monitoring Platform. The developed system successfully transformed athlete physiological measurements into interactive analytical outputs, evidence-based risk classifications, and AI-assisted health interpretations. The results demonstrate the successful integration of Microsoft Power BI, Python, and the Gemini 2.5 Flash model within a unified athlete monitoring environment.

The implemented platform produced six analytical dashboard modules that supported athlete monitoring at both individual and population levels. The generated outputs included physiological visualizations, risk indicators, comparative analytics, trend monitoring, evidence-based risk classification results, and AI-generated health interpretations based on HR, SpO_2, and Body Temperature measurements.

Athlete Health Overview Results

The Athlete Health Overview dashboard provided a high-level representation of overall team physiological status. The results displayed the total number of monitored athletes, team risk percentage, highest-risk athlete identification, critical athlete indicators, and physiological stress trends. The dashboard successfully highlighted athletes requiring attention through color-coded health classifications and dynamic KPI indicators, enabling rapid identification of abnormal physiological conditions.

Figure 18Athlete Health Overview Results

Detailed Analytics Results

The Detailed Analytics dashboard generated population-level analytical insights through risk distribution charts, monitoring coverage indicators, scatter visualizations, and athlete monitoring tables. The results allowed users to explore physiological patterns across multiple athletes and evaluate the distribution of health-risk categories within the monitored population.

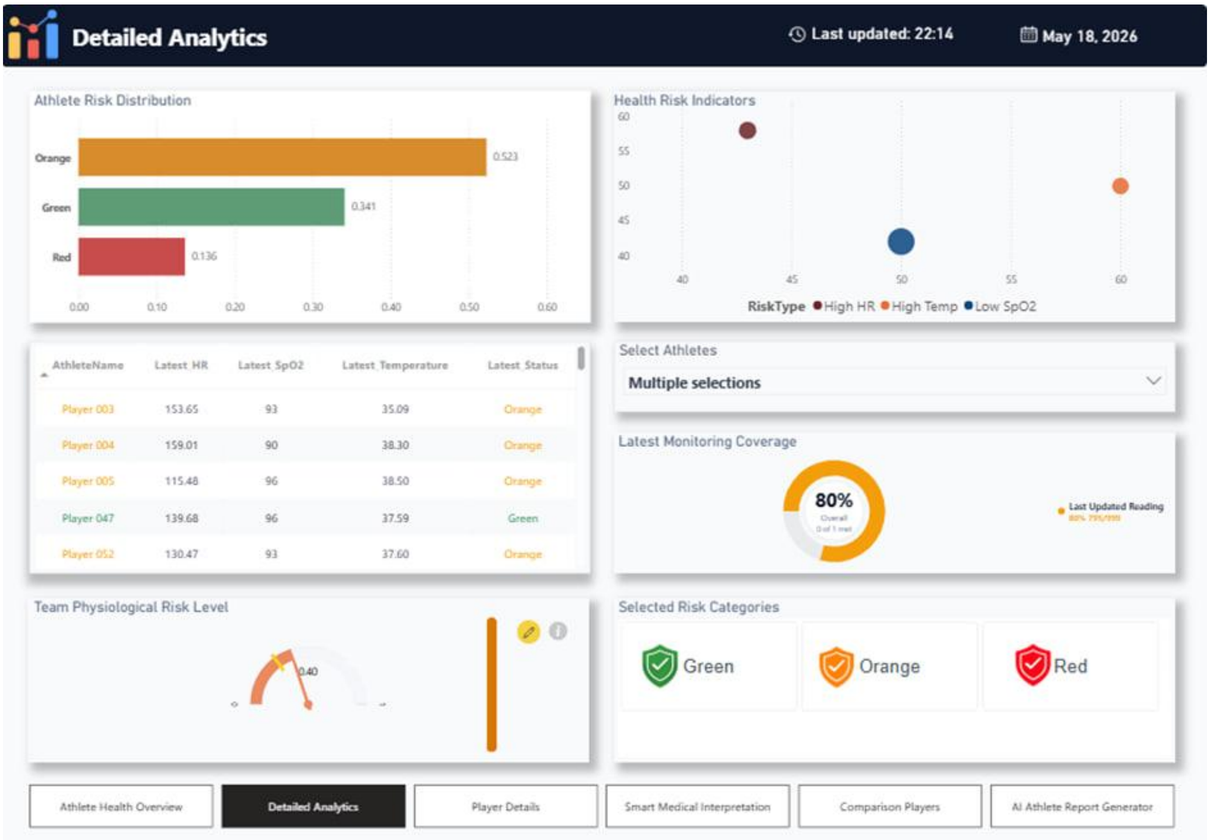

Figure 19Detailed Analytics Results

Player Details Results

The Player Details dashboard provided athlete-specific monitoring results by displaying detailed physiological measurements, health status indicators, risk percentages, trend visualizations, and monitoring alerts. The implemented dashboard dynamically updated according to athlete selection and successfully supported individual athlete assessment and longitudinal monitoring.

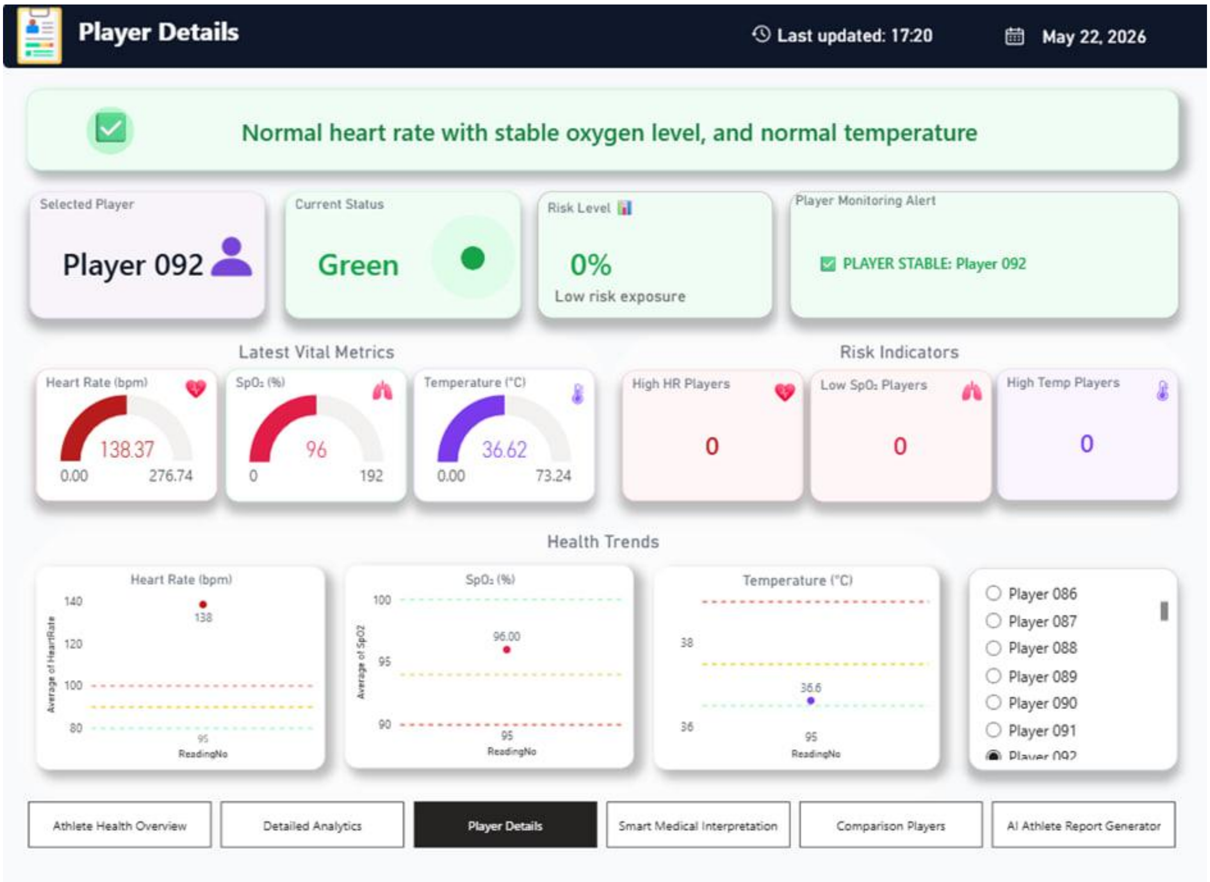

Figure 20Player Details Results

Smart Medical Interpretation Results

The Smart Medical Interpretation dashboard produced structured physiological assessments by combining classification outputs, diagnostic indicators, and risk contribution analysis. The generated results identified the physiological variables contributing most significantly to athlete risk conditions and provided recommendation-oriented monitoring insights. The dashboard successfully translated numerical measurements into understandable analytical summaries suitable for coaches and medical teams.

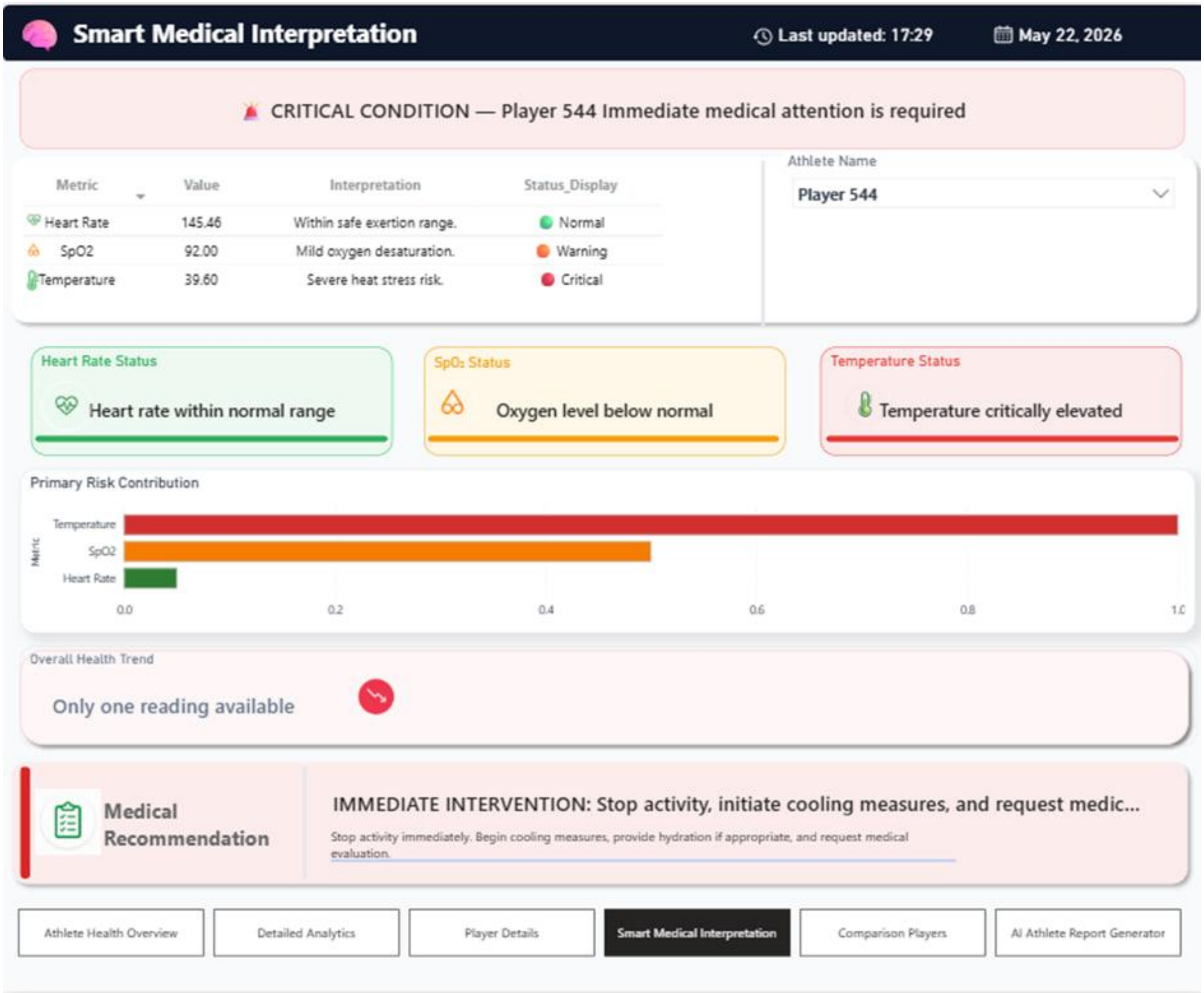

Figure 21Smart Medical Interpretation Results

Comparison Players Results

The Comparison Players dashboard generated side-by-side physiological comparisons between selected athletes. The results included comparative metric cards, radar visualizations, risk evaluations, and recommendation summaries. This functionality enabled efficient identification of physiological differences and supported comparative performance assessment within a unified analytical view.

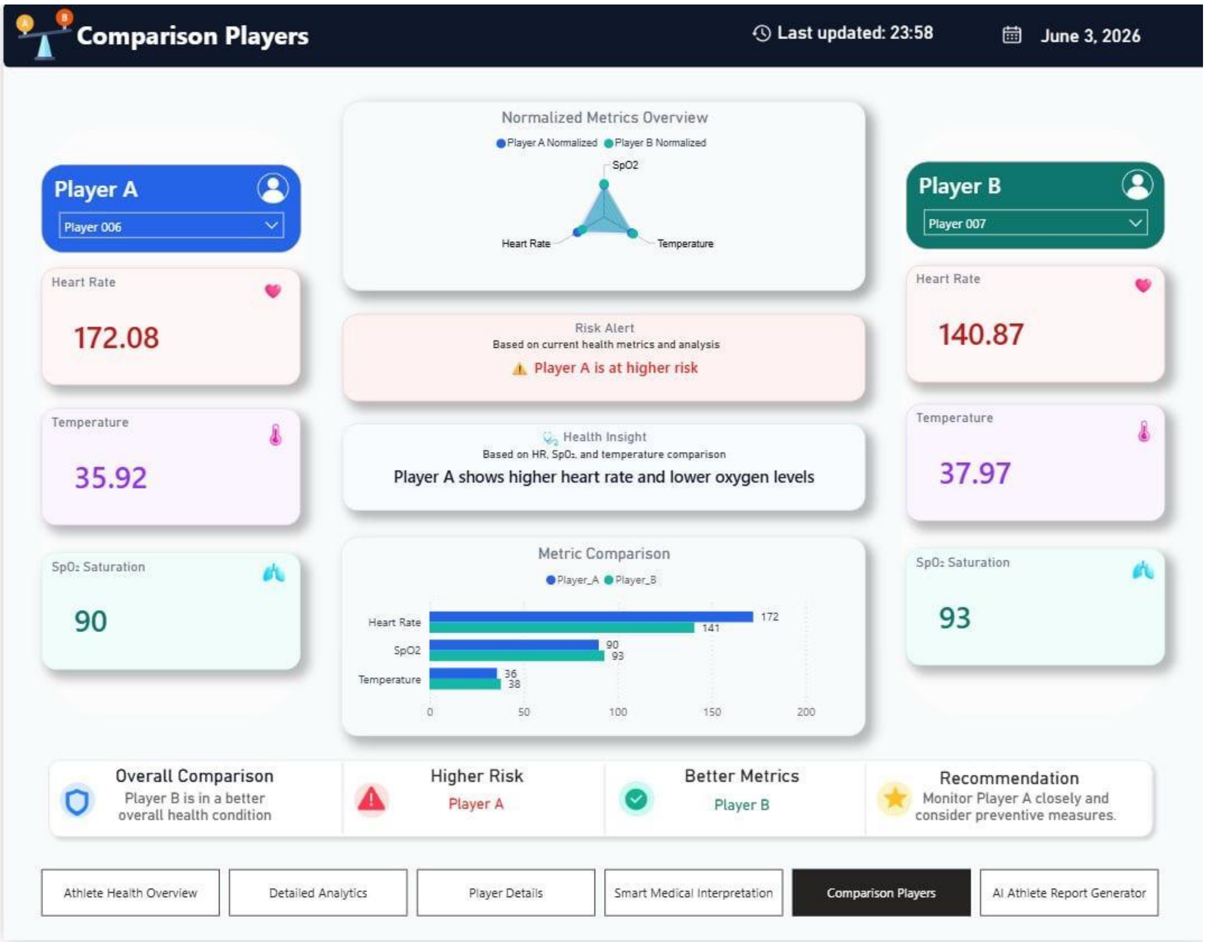

Figure 22Comparison Players Results

AI Athlete Report Generator Results

One of the most significant outcomes of the project was the successful implementation of the AI Athlete Report Generator. Through Python integration and Gemini 2.5 Flash LLM, the system automatically generated structured athlete health interpretation reports based on physiological measurements and classification results. The generated reports contained athlete condition summaries, key physiological findings, risk-oriented observations, monitoring insights, and recommendation-based outputs. These results demonstrated the ability of the system to transform complex physiological data into concise and understandable analytical interpretations.

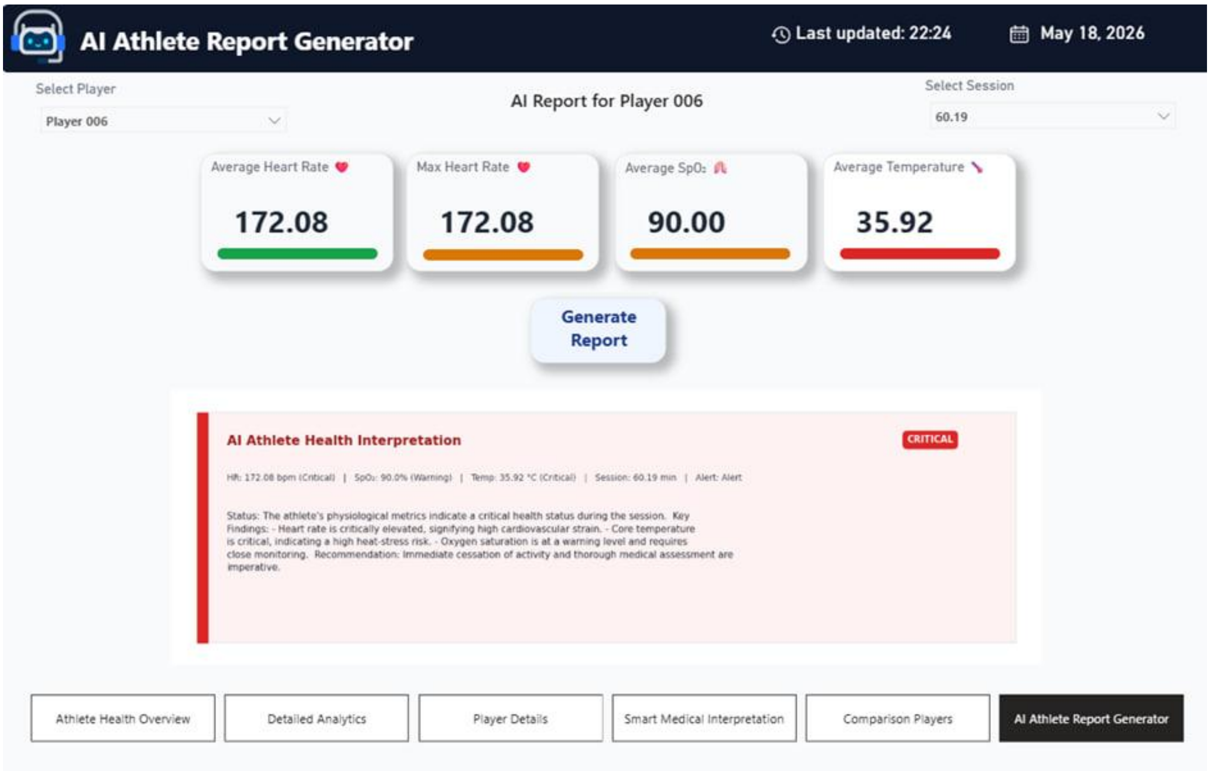

Figure 23AI Athlete Report Generator Results

In addition to dashboard visualizations, the implemented evidence-based classification framework successfully categorized athlete conditions into Green (Normal), Orange (Warning), and Red (Critical) risk levels using predefined physiological thresholds derived from scientific literature. The classification results were consistently reflected across KPI indicators, dashboard visuals, alerts, trend analyses, and AI-generated reports.

Overall, the results demonstrate that the developed platform successfully achieved the project objectives by integrating physiological monitoring, interactive analytics, risk classification, and AI-assisted interpretation within a single intelligent environment. The system provided meaningful insights that support athlete health monitoring, risk identification, and data-driven decision-making.

7.2 Functional Testing Results

Functional testing was conducted to verify that all major components of the VYTAL Athlete Health Monitoring Platform operated correctly and produced the expected outputs according to the system requirements. The testing process focused on validating dashboard functionality, physiological data processing, AI-based health classification, interactive visualization behavior, player comparison features, and AI-powered report generation.

A series of test scenarios were executed using different athlete records representing normal, warning, and critical physiological conditions. Each test case evaluated whether the system responded correctly to user actions, analytical calculations, classification rules, and AI interpretation requests. The testing results confirmed that the implemented platform successfully satisfied the functional requirements defined during the analysis and design phases.

Test ID	Function Tested	Expected Result	Actual Result	Status
FT-01	Load Athlete Dataset	Athlete data loads successfully into dashboard environment	Dataset loaded correctly and dashboard initialized successfully	Pass
FT-02	Display Physiological Measurements	HR, SpO_2, and Temperature values displayed correctly	Physiological measurements displayed accurately for all athletes	Pass
FT-03	Athlete Selection and Filtering	Dashboard updates according to selected athlete	All visual components updated dynamically after athlete selection	Pass
FT-04	Session Filtering	Dashboard displays selected session information correctly	Session-based filtering worked successfully across all pages	Pass
FT-05	Evidence-Based Risk Classification	Athlete condition classified according to threshold rules	Green, Orange, and Red classifications generated correctly	Pass
FT-06	Risk Indicator Visualization	Dashboard displays appropriate risk indicators and alerts	Risk indicators updated correctly based on classification results	Pass
FT-07	KPI Calculation Accuracy	KPI cards display correct analytical values	Calculated values matched underlying dataset and DAX outputs	Pass
FT-08	Trend Analysis Visualization	Trend charts update dynamically according to selected data	Trend visualizations updated correctly without inconsistencies	Pass
FT-09	Player Comparison Module	Comparison dashboard displays selected athletes correctly	Comparative metrics and charts generated successfully	Pass
FT-10	Risk Contribution Analysis	System identifies major physiological risk contributors	Risk contribution visuals displayed correct analytical outputs	Pass
FT-11	Smart Medical Interpretation Dashboard	Dashboard generates structured physiological assessment	Analytical summaries and recommendations displayed correctly	Pass
FT-12	AI Athlete Report Generation	Gemini generates structured athlete health interpretation	AI reports generated successfully with relevant findings and recommendations	Pass
FT-13	Dashboard Navigation	Users can navigate between dashboard modules smoothly	Navigation tabs functioned correctly across all pages	Pass
FT-14	Cross-Page Synchronization	Filters and selections remain synchronized between dashboards	Dashboard interactions remained consistent across pages	Pass
FT-15	Exception Handling During AI Requests	System handles unavailable AI responses appropriately	Fallback analytical summaries displayed successfully	Pass

Table 7 Functional Testing Results

The testing results demonstrated that all major functionalities operated as intended without critical execution failures. Dashboard components responded correctly to user interactions, analytical calculations remained consistent across different scenarios, and the evidence-based classification framework accurately categorized athlete conditions according to the implemented physiological thresholds.

Special attention was given to validating the AI-powered interpretation workflow because it represents one of the most advanced components of the platform. Multiple athlete scenarios covering Green, Orange, and Red risk categories were tested to verify that the generated interpretations remained consistent with the underlying physiological measurements and classification outputs. The results confirmed that the generated reports accurately reflected athlete conditions and provided structured interpretation outputs and monitoring recommendations consistent with the underlying physiological measurements.

Furthermore, interaction testing confirmed that filters, slicers, athlete selection controls, and navigation components remained synchronized across all dashboard pages. Visualizations, KPI indicators, risk alerts, comparison modules, and AI-generated outputs updated dynamically without calculation conflicts or data inconsistencies.

Overall, the functional testing process confirmed that the VYTAL platform successfully met its functional requirements and provided a reliable environment for athlete physiological monitoring, evidence-based risk classification, interactive analytics, and AI-assisted health interpretation.

7.3 Performance Evaluation

The performance of the VYTAL Athlete Health Monitoring Platform was evaluated to assess its efficiency, scalability, and reliability within the implemented analytical environment. The evaluation focused on measuring the responsiveness of dashboard interactions, the effectiveness of analytical processing workflows, the stability of the AI-assisted interpretation process, and the system's ability to support future expansion without significant architectural modifications.

Efficiency Evaluation

The developed platform demonstrated efficient analytical performance throughout the testing and deployment phases. Microsoft Power BI successfully processed physiological measurements, analytical calculations, and classification outputs while maintaining responsive dashboard interactions. User actions such as athlete selection, session filtering, dashboard navigation, and comparative analysis triggered dynamic updates across visual components without noticeable delays.

The implemented relational data model contributed significantly to analytical efficiency by organizing physiological data into structured tables connected through optimized relationships. In addition, DAX measures were designed to minimize unnecessary calculations while supporting dynamic KPI generation, trend analysis, risk classification, and comparative analytics.

The AI-powered interpretation workflow also demonstrated satisfactory performance. Selected athlete physiological measurements were processed through the Python integration layer and transmitted to the Gemini 2.5 Flash LLM for report generation. The generated interpretations were returned and displayed within the dashboard environment in a reasonable response time, providing users with automated analytical insights without disrupting dashboard usability.

Overall, the implemented analytical workflow successfully transformed physiological measurements into meaningful visualizations, classifications, and AI-generated reports while maintaining efficient system responsiveness.

Scalability Evaluation

The system architecture was designed using a modular and scalable analytical structure that supports future expansion and enhancement. The implemented data model separates physiological measurements, analytical calculations, classification logic, dashboard visualization, and AI-processing components into independent modules, allowing additional functionalities to be integrated without major modifications to the existing architecture.

The platform can be extended to accommodate additional physiological indicators, monitoring variables, or analytical models by expanding the relational data model and updating the classification framework. Similarly, new dashboard pages, analytical measures, and visualization components can be incorporated into the Power BI environment while preserving the existing system structure.

Furthermore, the AI integration layer was developed independently from the dashboard visualization layer, enabling future replacement or enhancement of the language model without affecting the remaining analytical components. This modular design improves maintainability and supports long-term system evolution.

The scalability evaluation indicates that the developed platform provides a flexible analytical foundation capable of supporting future enhancements such as wearable device integration, real-time data streaming, predictive analytics, and advanced athlete performance monitoring.

Reliability Evaluation

System reliability was evaluated through repeated execution of different monitoring scenarios representing Green, Orange, and Red physiological conditions. The platform consistently produced accurate analytical outputs, stable dashboard behavior, and reliable classification results across multiple test cases.

The AI-based health classification framework demonstrated reliable operation by consistently applying evidence-based physiological thresholds to athlete measurements and generating appropriate risk classifications. Dashboard indicators, KPI cards, trend charts, and risk visualizations remained synchronized with the calculated classification outputs throughout testing.

The reliability of the AI-powered interpretation workflow was also evaluated through repeated report-generation scenarios. Generated reports remained aligned with the provided physiological measurements and reflected the corresponding risk classifications accurately. Exception-handling mechanisms further improved reliability by generating fallback analytical summaries whenever AI responses were unavailable or communication interruptions occurred.

In addition, dashboard filters, navigation controls, cross-page interactions, and comparative analytical features operated consistently without calculation conflicts, data synchronization issues, or visualization errors. This behavior confirmed the stability of the integrated analytical environment.

Performance Evaluation Summary

The performance evaluation was conducted through repeated execution of functional testing scenarios and dashboard interaction assessments. The results demonstrated stable operation of the analytical environment across different athlete monitoring scenarios. Dashboard visualizations, KPI calculations, filtering operations, classification outputs, and AI-assisted interpretation workflows remained operational throughout testing without critical failures or synchronization issues.

The modular architecture implemented using Power BI, Python, and Gemini integration also demonstrated flexibility for future enhancement and expansion. In addition, repeated testing confirmed the consistency of classification outputs, dashboard interactions, and AI-generated interpretations across multiple physiological risk scenarios.

Overall, the evaluation indicates that the developed platform provides a stable, reliable, and maintainable analytical environment capable of supporting athlete physiological monitoring, evidence-based risk classification, and AI-assisted interpretation.

Evaluation Aspect	Observation
Dashboard Functionality	Operated correctly during testing
Analytical Calculations	Produced consistent outputs
Risk Classification	Applied threshold rules correctly
AI Report Generation	Generated structured interpretations successfully
Reliability	Consistent across multiple test scenarios

Table 8Performance Evaluation Summary

7.4 Usability Testing / User Feedback

Usability assessment was conducted to evaluate the effectiveness, accessibility, and overall user experience of the VYTAL Athlete Health Monitoring Platform. The objective of this evaluation was to determine whether the implemented dashboard interface enabled efficient athlete monitoring, interpretation of analytical results, navigation between dashboard modules, and utilization of AI-generated insights with minimal complexity.

The assessment focused on the main dashboard components, including the Athlete Health Overview, Detailed Analytics, Player Details, Smart Medical Interpretation, Comparison Players, and AI Athlete Report Generator modules. Several interaction scenarios were performed to assess dashboard navigation, information readability, visualization clarity, filtering functionality, and interpretation of health-risk indicators.

During the assessment process, different athlete monitoring scenarios were executed by selecting athletes, filtering monitoring sessions, reviewing physiological measurements, exploring trend visualizations, comparing athletes, and generating AI-powered health reports. Observations were collected regarding the ease of completing monitoring tasks, the clarity of dashboard outputs, and the effectiveness of the overall interface design.

The assessment results indicated that the dashboard interface was intuitive and easy to navigate. Users were able to move between analytical modules efficiently using the implemented navigation structure without requiring extensive guidance. The modular organization of dashboard pages helped users locate relevant information quickly and reduced the effort required to analyze athlete conditions.

The implemented color-coded classification framework was identified as one of the most effective usability features. The consistent use of Green, Orange, and Red status indicators enabled rapid recognition of athlete health conditions and supported efficient identification of higher-risk cases. This visual approach improved situational awareness and reduced the time required to interpret physiological measurements.

The implemented KPI cards, physiological gauges, trend visualizations, and risk indicators provided clear summaries of athlete conditions and simplified the interpretation of physiological data. The structured dashboard layout and organized analytical sections improved readability and reduced visual clutter during interaction.

The AI Athlete Report Generator enhanced the interpretability of physiological measurements by automatically transforming numerical values into structured health interpretations and monitoring recommendations. The generated reports provided concise summaries that supported analytical review and decision-making within the dashboard environment.

In addition, the interactive filtering and athlete-selection features demonstrated effective usability. Dashboard visualizations updated dynamically according to user selections, allowing efficient exploration of athlete data without requiring page reloads or additional processing steps. This responsiveness improved the overall analytical experience and supported smooth interaction with the monitoring environment.

Table 9Usability Evaluation Summary

Evaluation Aspect	Observation
Dashboard Navigation	Easy and intuitive navigation between modules
Information Readability	Physiological information displayed clearly and consistently
Risk Visualization	Color-coded classifications improved rapid interpretation
Dashboard Responsiveness	Visualizations updated dynamically and efficiently
Athlete Monitoring	Users could identify athlete conditions quickly
AI Report Interpretation	Generated reports improved understanding of physiological data
User Interaction	Filters and slicers were easy to use and responsive
Overall User Experience	Positive and supportive for analytical decision-making

Overall, the usability assessment demonstrated that the VYTAL platform provides an effective and user-friendly monitoring environment capable of supporting athlete health assessment, analytical exploration, and AI-assisted decision-making. The combination of interactive visualizations, structured dashboard layouts, dynamic filtering, and AI-generated interpretations contributed to an intuitive user experience and enhanced the practical usability of the developed system.

7.5 Comparative Analysis

To further evaluate the effectiveness of the VYTAL Athlete Health Monitoring Platform, a comparative analysis was conducted between the developed system and existing monitoring approaches reported in the literature. The objective of this comparison was to determine how the implemented platform addresses common limitations identified in previous studies and to evaluate the additional capabilities introduced through the integration of interactive analytics, evidence-based risk classification, and AI-assisted interpretation.

As discussed in Chapter 2, many athlete monitoring solutions focus primarily on collecting physiological measurements and presenting them through basic visualizations. While these approaches provide useful monitoring functionality, they often require manual interpretation of physiological readings and may lack integrated analytical and decision-support capabilities.

The VYTAL platform was developed to address these limitations by combining physiological monitoring, evidence-based risk classification, interactive analytics, and AI-assisted interpretation within a single integrated environment. Unlike conventional monitoring approaches that rely primarily on manual evaluation, the developed system automatically categorizes athlete conditions into Green, Orange, and Red risk levels based on scientifically validated physiological thresholds for HR, SpO_2, and Body Temperature.

Table 10Comparative Analysis of Athlete Monitoring Systems

Feature	Existing Monitoring Approaches	VYTAL Platform
Physiological Data Monitoring	Available	Available
Interactive Dashboard Visualization	Available	Available
Evidence-Based Risk Classification	Limited	Available
Risk Contribution Analysis	Limited	Available
Athlete Comparison Functionality	Limited	Available
AI-Assisted Interpretation	Rarely Reported	Available
AI-Assisted Recommendations	Rarely Reported	Available
Integrated Decision Support	Limited	Available

The comparison results demonstrate that the VYTAL platform provides a more comprehensive analytical environment than existing monitoring approaches reported in the literature.

In addition to visualizing physiological measurements, the system performs evidence-based risk classification, generates analytical insights, supports athlete comparison, and delivers AI-assisted interpretations that enhance monitoring efficiency and decision-making capabilities.

Overall, the comparative analysis confirms that the developed system extends the capabilities of conventional athlete monitoring approaches by introducing intelligent analytical functions, automated health assessment, and AI-assisted interpretation. These enhancements contribute to improved situational awareness, more efficient athlete monitoring, and stronger support for data-driven decision-making within sports-health environments.

7.6 Discussion of Findings

The findings obtained from the implementation and evaluation phases demonstrate that the VYTAL Athlete Health Monitoring Platform successfully achieved the objectives defined at the beginning of the project. The developed system was able to transform raw physiological measurements into meaningful analytical insights through the integration of interactive visualization, evidence-based health classification, and automated interpretation mechanisms.

One of the most significant findings was the effectiveness of combining Microsoft Power BI and artificial intelligence technologies within a unified analytical environment. The implemented dashboards successfully presented physiological measurements in a structured and understandable manner, allowing users to monitor athlete conditions, identify abnormal physiological patterns, and evaluate health-risk indicators efficiently. The results confirmed that visual analytics can significantly improve the interpretation of physiological data compared with conventional approaches that rely primarily on numerical measurements and manual assessment.

The implemented evidence-based health classification framework also proved to be a valuable component of the system. By applying scientifically validated physiological thresholds to HR, SpO_2, and Body Temperature measurements, the platform was able to automatically categorize athlete conditions into Green, Orange, and Red risk levels. The generated classifications remained consistent throughout different testing scenarios and enabled rapid identification of athletes requiring additional monitoring or medical attention. This finding highlights the importance of automated risk assessment in supporting proactive athlete health management.

Another important finding was the contribution of the Gemini Large Language Model to improving data interpretability. Conventional monitoring approaches often require users to manually analyze physiological measurements and determine their significance. In contrast, the developed platform automatically generated structured health interpretations and recommendation-oriented reports based on athlete physiological conditions. The generated reports transformed complex numerical data into understandable analytical summaries, reducing interpretation effort and improving decision-support capabilities for coaches and medical teams.

The evaluation results further demonstrated the effectiveness of interactive dashboard technologies in supporting athlete monitoring activities. Dynamic filtering, athlete selection, trend analysis, risk visualization, and comparison functionalities enabled users to explore physiological data efficiently and obtain meaningful insights from different analytical perspectives. These features contributed to improved situational awareness and supported data-driven decision-making processes within the monitoring environment.

The findings also support the research gap identified in Chapter 2. Previous studies primarily focused on dashboard visualization, decision-support analytics, athlete monitoring, or large language models as separate technological domains. The results of this project demonstrate that integrating these technologies within a single intelligent ecosystem provides greater analytical value than using each component independently. The combination of interactive analytics, automated classification, and AI-generated interpretation created a comprehensive decision-support environment capable of enhancing athlete monitoring efficiency and improving the overall user experience.

Furthermore, the usability assessment indicated that the dashboard interface supported effective interaction and interpretation of analytical outputs with minimal difficulty. The consistent dashboard structure, intuitive navigation, and color-coded risk indicators improved readability and simplified the monitoring process. This finding emphasizes the importance of user-centered design when developing analytical systems intended for practical decision-support applications.

Overall, the findings confirm that the VYTAL platform successfully addressed the challenges associated with athlete physiological monitoring by providing intelligent analytical capabilities, automated risk assessment, and AI-assisted interpretation within a unified monitoring environment. The significance of these findings lies in demonstrating how the integration of artificial intelligence, business intelligence, and evidence-based analytical methods can enhance athlete health assessment, improve monitoring efficiency, and support more informed decision-making in sports-health environments.

7.7 Challenges Encountered and Lessons Learned

During the development and evaluation of the VYTAL Athlete Health Monitoring Platform, several technical and analytical challenges were encountered. These challenges were addressed through iterative testing, refinement, and optimization of the implemented system components. The experience gained from solving these challenges contributed to improving the reliability, usability, and analytical quality of the final platform.

One of the main challenges was preparing the athlete physiological dataset for accurate analysis. The original dataset required cleaning, restructuring, and validation before it could be used effectively within the dashboard environment. Inconsistent values, duplicated records, and formatting differences could affect the accuracy of analytical calculations and dashboard outputs. This challenge was addressed by applying data preprocessing procedures using Power Query and Python, including duplicate removal, data transformation, physiological value validation, and organization of records into structured analytical tables. This process highlighted the importance of data quality as a foundation for reliable analytical systems.

Another challenge was implementing accurate health classification logic based on multiple physiological indicators. The system needed to evaluate HR, SpO_2, and Body Temperature simultaneously and classify athlete conditions into Green, Orange, and Red risk levels. Ensuring that the classification rules were applied consistently across KPI cards, charts, tables, and AI-generated reports required careful design of DAX calculations and repeated validation. This challenge was addressed by using evidence-based physiological thresholds and testing different athlete scenarios to confirm that the classification outputs matched the expected risk levels.

Dashboard interaction and cross-page synchronization also represented an important challenge. Since the platform included multiple dashboard pages, such as Athlete Health Overview, Detailed Analytics, Player Details, Smart Medical Interpretation, Comparison Players, and AI Athlete Report Generator, user selections and filters needed to update visual components consistently. This challenge was addressed by refining table relationships, testing slicers and filters, validating cross-page interactions, and ensuring that all analytical visuals were connected to the correct data model and calculated measures.

The integration of Python with the Gemini Large Language Model introduced additional implementation challenges. The AI report generation process required extracting selected athlete measurements from Power BI, preparing structured prompts, communicating with the Gemini API, processing generated responses, and rendering the final interpretation inside the dashboard. Variations in AI-generated responses and occasional API availability issues could affect report generation. This challenge was addressed by applying controlled prompt-engineering techniques and implementing fallback analytical summaries to maintain system reliability when AI responses were unavailable.

Another challenge was designing the dashboard interface in a way that balanced analytical detail with usability. The system needed to present several physiological indicators, risk metrics, comparison visuals, and AI-generated reports without overwhelming users. This challenge was addressed by adopting a modular dashboard layout, using consistent color-coded classifications, organizing KPI cards and charts clearly, and separating each analytical objective into a dedicated dashboard page. This improved readability and reduced visual complexity.

Several lessons were learned throughout the project development process. First, data preprocessing is one of the most critical stages in building reliable analytical dashboards, as inaccurate or inconsistent data can affect all subsequent calculations and visualizations. Second, evidence-based classification systems require repeated testing across different scenarios to ensure that analytical rules remain consistent and interpretable. Third, integrating AI-generated interpretation into analytical dashboards can significantly improve data understanding, but it requires careful prompt design, response validation, and exception handling. Finally, user interface design plays an important role in the success of decision-support systems because clear visual organization directly affects the user's ability to interpret results and make informed decisions.

Overall, the challenges encountered during the project contributed to improving the final implementation of the VYTAL platform. Addressing these challenges enhanced data quality, strengthened the reliability of the evidence-based classification framework, improved dashboard usability, and increased the robustness of the AI-assisted interpretation workflow. The experience gained throughout the development process provided valuable insights into the design and implementation of intelligent monitoring systems that combine business intelligence, physiological analytics, and artificial intelligence within a unified analytical environment.

7.8 Limitations

Although the VYTAL Athlete Health Monitoring Platform successfully achieved its intended objectives, several limitations should be acknowledged. Identifying these limitations is important for understanding the current scope of the system and for guiding future enhancements and research efforts.

One of the primary limitations of the project is the reliance on a pre-existing dataset obtained from Kaggle rather than real-time physiological measurements collected directly from athletes. While the dataset provided sufficient information for developing, testing, and validating the analytical framework, it does not fully represent the dynamic nature of live athlete monitoring environments. Consequently, the current implementation operates within a simulated monitoring setting rather than a real-time deployment scenario.

Another limitation is the use of a rule-based health classification framework based on predefined physiological thresholds. Although these thresholds were derived from scientifically validated literature and provide a structured foundation for risk assessment, they may not account for individual athlete characteristics such as age, fitness level, medical history, sport type, or physiological adaptation. As a result, the classification system applies generalized thresholds rather than personalized athlete-specific evaluations.

The AI-generated interpretation component also has certain limitations. The Gemini Large Language Model was integrated to generate analytical summaries and interpretation reports and monitoring recommendations based on athlete physiological measurements. While the generated outputs improved data interpretability and decision support, they remain dependent on the quality of the provided input data and the generated prompt structure. Furthermore, the AI-generated reports are intended to support monitoring and analytical understanding rather than provide clinical diagnoses or replace professional medical judgment.

In addition, the current system focuses primarily on three physiological indicators: HR, SpO_2, and Body Temperature. Although these indicators are important measures of athlete condition and physiological stress, other relevant variables such as blood pressure, respiratory rate, sleep quality, hydration status, workload metrics, and injury-related indicators were not included within the current project scope. Incorporating additional physiological parameters could further improve monitoring accuracy and analytical depth.

Another limitation relates to external service dependency for AI-powered report generation. The AI interpretation workflow relies on communication with the Gemini API, meaning that report generation performance may be influenced by internet connectivity, service availability, API limitations, or response delays. Although mitigation strategies were implemented to improve reliability, dependence on external AI services remains an inherent limitation of the current architecture.

Finally, the platform was designed primarily as a decision-support and analytical monitoring system rather than a predictive healthcare solution. The implemented framework focuses on visualizing current athlete conditions, performing risk classification, and generating analytical interpretations. Advanced predictive capabilities, such as injury prediction, fatigue forecasting, anomaly detection, and personalized performance optimization models, were beyond the scope of the current implementation.

Despite these limitations, the developed platform successfully demonstrates the feasibility and value of integrating business intelligence, artificial intelligence, and evidence-based physiological analytics within a unified athlete health monitoring environment. The identified limitations also provide clear opportunities for future enhancement and expansion of the system.

Chapter 8

Conclusion and Future Work

8.1 Summary of Work

This project focused on the design and development of the VYTAL Athlete Health Monitoring Platform, an intelligent system that integrates physiological data analytics, interactive visualization, evidence-based risk classification, and AI-assisted interpretation to support athlete health monitoring and decision-making.

The project began with a review of existing literature related to healthcare dashboards, athlete monitoring systems, decision-support technologies, and Large Language Models. The literature review identified a gap in the integration of physiological monitoring, analytical visualization, automated risk assessment, and AI-generated interpretation within a unified platform.

To address this gap, a modular system architecture was designed and implemented using Microsoft Power BI, Python, and the Gemini Large Language Model. The development process included data preprocessing, data transformation, relational data modeling, dashboard development, evidence-based classification design, Python integration, and AI-assisted report generation.

The implemented platform analyzed key physiological indicators, including HR, SpO_2, and Body Temperature. Physiological measurements were processed through an evidence-based classification framework that categorized athlete conditions into Green, Orange, and Red risk levels. The generated results were presented through multiple interactive dashboard modules that supported athlete monitoring, trend analysis, player comparison, risk assessment, and health interpretation.

The system was subsequently evaluated through functional testing, performance assessment, usability evaluation, and comparative analysis. The evaluation results demonstrated that the developed platform operated successfully and provided an integrated environment for athlete physiological monitoring, analytical exploration, evidence-based risk assessment, and AI-assisted interpretation.

8.2 Key Findings

The results obtained from the development and evaluation of the VYTAL Athlete Health Monitoring Platform revealed several important findings. First, the integration of Microsoft Power BI, Python, and the Gemini Large Language Model successfully created a unified environment for athlete health monitoring, analytical visualization, and AI-assisted interpretation.

Second, the implemented evidence-based classification framework effectively categorized athlete conditions into Green, Orange, and Red risk levels using scientifically validated physiological thresholds for HR, SpO_2, and Body Temperature. The classification results remained consistent across different testing scenarios and supported rapid identification of athletes requiring additional attention.

Third, the interactive dashboard environment improved the accessibility and interpretability of physiological data. Features such as KPI indicators, trend analysis, athlete comparison, risk visualization, and monitoring summaries enabled efficient exploration of athlete health conditions from multiple analytical perspectives.

Another key finding was the successful integration of the Gemini Large Language Model for generating structured health interpretations and monitoring recommendations. The AI-generated reports transformed physiological measurements into concise analytical summaries, improving the interpretability of athlete data and supporting decision-making processes.

The evaluation results also demonstrated that the developed platform operated reliably across multiple testing scenarios. Dashboard interactions, analytical calculations, classification outputs, and AI-assisted interpretation workflows functioned consistently without major operational issues.

Overall, the project demonstrated the feasibility and value of integrating business intelligence, evidence-based physiological analytics, and artificial intelligence within a single athlete health monitoring platform. The findings indicate that such integration can enhance athlete monitoring, improve situational awareness, and support more informed data-driven decision-making in sports-health environments.

8.3 Limitations and Future Work

Although the VYTAL Athlete Health Monitoring Platform successfully achieved its objectives, several limitations were identified during the development and evaluation process. The current implementation relies on a pre-existing dataset rather than real-time physiological measurements collected directly from athletes. In addition, the evidence-based classification framework applies generalized physiological thresholds and does not incorporate athlete-specific characteristics such as age, fitness level, sport type, or medical history. The AI-assisted interpretation component is also dependent on the quality of the input data and the availability of external AI services. Furthermore, the system currently focuses on three physiological indicators HR, SpO_2, and Body Temperature and does not include additional monitoring parameters that may provide deeper insight into athlete health and performance.

These limitations create several opportunities for future development and research. One potential enhancement is the integration of wearable sensors and IoT devices to enable real-time physiological monitoring and continuous data collection. Future versions of the platform could also incorporate additional physiological and performance-related indicators, such as blood pressure, respiratory rate, sleep quality, hydration status, workload metrics, and recovery measurements.

Another important direction for future work is the implementation of predictive analytical models capable of identifying injury risk, fatigue progression, abnormal physiological patterns, and performance decline before they occur. Machine learning techniques could be integrated to provide personalized athlete assessments based on historical physiological data and individual characteristics.

Further enhancements may also include the development of mobile applications for remote monitoring, cloud-based data integration, and advanced AI-assisted recommendation systems that provide more personalized monitoring insights. In addition, future research could evaluate the platform using real-world athlete data collected from sports organizations and training environments to further validate its effectiveness and practical applicability.

Overall, the identified limitations and future development opportunities provide a clear pathway for extending the capabilities of the VYTAL platform and advancing the use of intelligent technologies in athlete health monitoring and decision-support systems.

Appendix A

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Appendix B

This appendix provides access to the complete VYTAL Athlete Health Monitoring Platform developed as part of this project. The Power BI workspace contains all implemented dashboard modules, including the Athlete Health Overview, Player Details, Risk Profiling, AI-Based Health Interpretation, Comparative Analytics, and supporting visualizations developed throughout the project.

The dashboard can be accessed through the following Google Drive repository:

https://drive.google.com/drive/folders/15zuQ0rzXbYVCKdlOtwlBMXaq61E-eTKT

The repository includes the Power BI project files, dashboard resources, supporting materials, and implementation artifacts used during the development and evaluation phases of the VYTAL Athlete Health Monitoring Platform.

Appendix C

The published Power BI dashboard for the VYTAL Athlete Health Monitoring Platform can be accessed using the following link:

https://app.powerbi.com/groups/me/reports/fcda16d3-f412-4593-a57f-87564df3e064/0f2c3402b56a420e48fa?experience=power-bi